
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is among the most popular neighbor embedding methods, and, like its contemporaries, it relies on attractive and repulsive forces among high-dimensional data points to build a low-dimensional embedding. This paper makes four contributions that together turn these forces from a black box into an interpretable design space. *First*, we recast the attractive and repulsive gradients as explicit *attraction* and *repulsion shapes* and derive exact conditions under which each contracts or expands distances. *Second*, using these shapes we explain a counterintuitive fact: attraction in UMAP can act simultaneously as attraction and repulsion in the low-dimensional map, which pins down precisely why learning-rate annealing is needed and why attractive and repulsive terms warrant different treatment. *Third*, we show that a targeted modification of the attraction shape measurably improves the consistency of cluster formation under random initialization, indicating that UMAP encodes a unique underlying structure. *Fourth*, we use the same shape lens to compare UMAP, Parametric UMAP, and NEG-*t*-SNE, exposing the similarities and differences that govern their stability. Taken together, these contributions make UMAP and similar embedding methods more interpretable, more robust, and more accurate.

1 Introduction

The current era is characterized by a deluge of high-dimensional data, and dimensionality reduction (DR) techniques have become essential tools for exploring such data by visualizing its underlying structure. The two most popular methods, *t*-distributed stochastic neighbor embedding (Maaten and Hinton [2008]) and uniform manifold approximation and projection (UMAP) (McInnes et al. [2018]), are both grounded in the same attraction-repulsion dynamics: similar data points are pulled closer while dissimilar ones are pushed apart. Because these are unsupervised algorithms, they use no labels; instead, they recover and preserve the intrinsic structure of high-dimensional data by leveraging local (attractive) and global (repulsive) relationships (forces). This label-free, data-driven design is exactly what makes them well-suited to tasks such as clustering (Becht et al. [2019]), exploratory data analysis (Fleischer and Islam [2020]), anomaly detection in semiconductor manufacturing (Fan et al. [2021]), visual search (González-Márquez et al. [2024]), time series analysis (Altin and Cakir [2024]), studying representation convergence (Huh et al. [2024]), and outlier image detection (Islam and Fleischer [2024]), where surfacing hidden patterns in unlabeled data is both critical and meaningful. In short, DR exemplifies the power of unsupervised methods to distill complex data into interpretable form—which is precisely why understanding the forces that drive it matters.

Building on this attraction-repulsion principle, a wave of newer methods has emerged (Amid and Warmuth [2019], Agrawal et al. [2021], Wang et al. [2021], Narayan et al. [2021], Yang et al. [2022], Wang et al. [2025], Kury et al. [2025]), each emphasizing specific aspects of the data. Yet despite their wide adoption, these methods lean on heuristic practices that can yield uninterpretable results, and the distortions DR introduces are unavoidable (Chari and Pachter [2023]). This gap motivates our work: a deeper, mechanistic understanding of the algorithms is needed so that practitioners can interpret embeddings correctly, avoid spurious structure, and tune performance deliberately. In practice, compact clusters are coaxed out through an assortment of tricks—careful initialization, learning-rate schedules, and kernel tuning—but the dynamics of the attractive and repulsive forces that actually produce clusters have never been studied thoroughly. Worse, the parameters that matter are buried inside abstract functional forms, which makes the algorithms hard to explain in the first place.

We address this gap by decomposing the forces into their constituent parts and extracting the functional *shapes* of attraction and repulsion. This single move lets us localize the source of three long-standing behaviors: the necessity of learning-rate annealing, the difficulty of producing consistent output under random initialization, and the origin of cluster formation all trace back to attraction, whereas repulsive forces primarily govern inter-cluster distances. Concretely, this paper makes the following contributions:

1. **A shape-based reformulation of the forces.** We recast the attractive and repulsive gradients as explicit attraction and repulsion shapes, establish exact conditions for the contraction and expansion of distance, and thereby provide a fresh, mechanistic perspective on this family of algorithms (Section 4).
2. **An explanation of learning-rate annealing.** We show that UMAP’s attraction shape produces the counter-intuitive behavior of both contracting and expanding distances, and, by comparing attraction shapes across algorithms, we pin down exactly how attraction dictates the need for a learning-rate annealing scheme (Section 5).
3. **A principled fix for initialization sensitivity.** We modify the attraction shape to improve the consistency of embeddings under random initialization, providing evidence that UMAP encodes a unique underlying structure (Section 5.1).
4. **A characterization of cluster formation.** By analyzing repulsion shapes, we deliver a deeper account of how clusters form and how inter-cluster distance is regulated (Section 5.2).
5. **A unifying comparison across methods.** We compare the attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, expose their similarities and distinctions, and use the shapes to characterize the stability of each algorithm (Section 6).

The main text centers on UMAP—now a de facto standard across many fields—and brings in other algorithms where they sharpen a point; the appendix supplies the corresponding details for each. The code used in this research will be released upon acceptance.

2 Related Works

The lineage of modern iterative graph-based neighbor embedding algorithms begins with stochastic neighbor embedding (SNE) (Hinton and Roweis, 2002) and its t-distribution-based extension, *t*-SNE (Maaten and Hinton, 2008). Both build a dense graph in which every point relates pairwise to all others—similar or not—and normalize the graph weights into a probability distribution. Concurrent methods, including locally linear embedding (Roweis and Saul, 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi, 2002), instead used a *k*-nearest neighbor (*k*-NN) graph of pairwise interactions; known as spectral methods, they rely on eigenvalue decomposition. The iterative and spectral threads were brought together by Tang et al. (Tang et al., 2016), who incorporated the *k*-NN graph into the iterative approach and dropped low-dimensional normalization. McInnes et al. (2018) pushed this further in UMAP, removing normalization altogether—in both high and low dimensions—and obtaining the embedding from pairwise interactions alone. Its optimization uses an explicit attractive force to preserve local neighborhoods and a repulsive force to keep dissimilar points apart; it is precisely these two forces that our shape reformulation dissects. Building on the same foundations, methods such as PaCMAP (Wang et al., 2021) and NEG-*t*-SNE (Damrich et al., 2023) have since been proposed. For a recent survey, see de Bodt et al. (2025).

Considerable effort has gone into understanding and explaining the relationships among these algorithms, and our contribution builds directly on this line of work. An early analysis of SNE established that data well-clustered in the original space remain well-clustered in the embedding (Shaham and Steinerberger, 2017). A parallel analysis of *t*-SNE by Linderman and Steinerberger (2019) showed that the number of clusters in the embedding lower-bounds the number in the original space, a result subsequently sharpened by further characterizations (Arora et al., 2018; Cai and Ma, 2022; Linderman and Steinerberger, 2022). Because *t*-SNE and UMAP share an underlying framework yet produce drastically different visualizations, a major thread has sought to connect them (Böhm et al., 2022; Damrich et al., 2023; Draganov et al., 2023). Böhm et al. (2022) theorized that Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al., 2014), UMAP, and *t*-SNE are all samples from a single underlying spectrum; LE and *t*-SNE are indeed linked through the early-exaggeration phase (Cai and Ma, 2022), and UMAP and *t*-SNE can be related through contrastive estimation (Damrich et al., 2023). Around the same time, Hu et al. (2023) independently uncovered the tie between contrastive learning and SNE. More recent work explains DR through a probabilistic lens (Ravuri et al., 2023; Ravuri and Lawrence, 2024), kernel techniques (Draganov and Dohn, 2023), and information geometry (Kolpakov and Rocke, 2024). Our shape-based account is complementary: rather than relating whole algorithms, it isolates the attractive and repulsive forces themselves as the objects to compare.

3 Uniform Manifold Approximation and Projection

UMAP constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ by using a pairwise relation: $p_{i,j} = f_h(d(x_i, x_j))$, typically, $\in [0, 1]$ (for details, see Appendix H).

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is given by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points.

This section fixes the notation and the two forces that the rest of the paper analyzes. UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

Here the first term supplies the attractive force and the second the repulsive force—the two ingredients our shape reformulation will separate. Rather than optimizing every point at each iteration, UMAP uses negative sampling (Mikolov et al. 2013; Tang et al. 2016): for each edge with $p_{ij} > 0$ (a positive edge), several edges are drawn at random (negative edges). The attractive force is applied on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force is applied on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number, and note that y_j is not updated for negative edges. This asymmetric update is the raw material for our first contribution, which repackages these gradients as shapes. For a detailed analysis of UMAP’s loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

Our first contribution is a reformulation that makes the forces analyzable. The updates (4, 6) simplify once we decompose the gradients $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_j^t} \log(1 - q_{ij})$) into a scalar coefficient—a function of the distance $\|y_i - y_j\|_2$ —acting on the vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) *shape*. We use UMAP as the running example, but the formalism applies to any method built on attraction and repulsion.

By writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, we can update the equations of a positive edge as

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and we use the fact that for Euclidean metric, $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. Similarly, we reformulate the update equation of a negative edge as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Decompositions of this kind have appeared before (McInnes et al. 2018; Agrawal et al. 2021; Draganov and Dohn 2023), but their purpose sets our contribution apart. The original UMAP paper (McInnes et al. 2018) used it as a computational trick for speed, and Draganov and Dohn (2023) used it to compare algorithms from a kernel viewpoint; in both, the goal was simply to compute the derivative. We instead treat the decomposed coefficients as *independent functions in their own right*, free to take various forms. Agrawal et al. (2021) took a related step but expressed the decomposition as a scalar coefficient times a unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to highlight force magnitude. As the analysis below demonstrates, our shape decomposition is more illuminating than magnitude alone—and it is what powers every subsequent contribution.

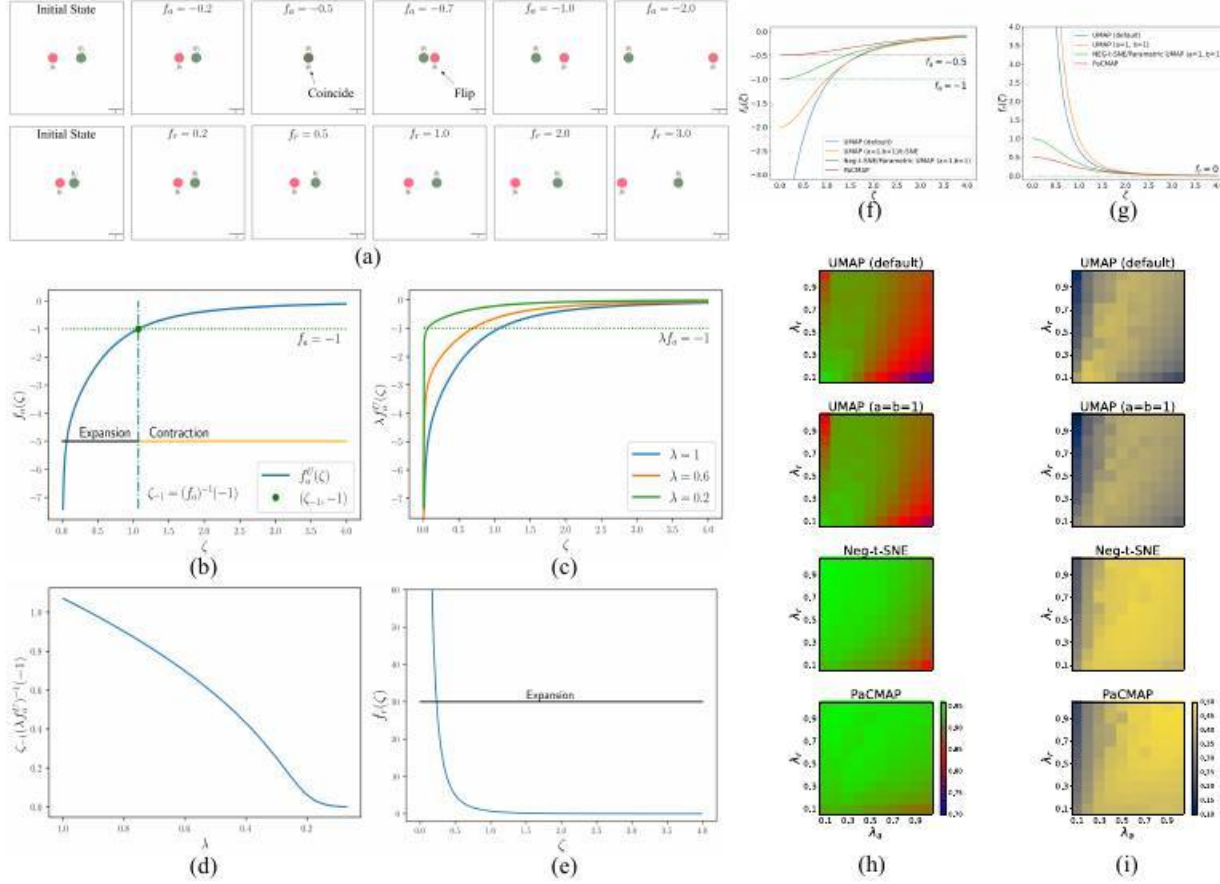


Figure 1: The attraction and repulsion shapes, our central analytical tool, at work in UMAP. (a) How different values of f_a (top) and f_r (bottom) move a pair. (b) UMAP’s attraction shape. (c) The effective attraction shape (λf_a) across learning rates λ . (d) The minimum distance for contraction (ζ_{-1}) shrinks as λ decreases. (e) UMAP’s repulsion shape. (f) Attraction and (g) repulsion shapes across embedding methods. (h) Trustworthiness and (i) Silhouette Score as the constant learning rate is varied independently for attraction (λ_a) and repulsion (λ_r). Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

Having defined the shapes, we now deliver the analytical core of the first contribution: exact conditions under which the updates contract or expand distances, stated directly in terms of the shapes. From the update equations (7)–(10), the following proposition characterizes the contraction of distance between the pair y_i^t and y_j^t of a positive edge:

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here, λf_a works as the effective attraction shape. In most cases, however, f_a alone is enough to draw meaningful conclusions about the embeddings. For a negative edge, the following proposition characterizes the expansion of the distance between the pair y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Note that adding a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ to Eq. (10) leaves this conclusion unchanged. The value of these two propositions is that they furnish a *per-iteration certificate* for guaranteed contraction or expansion. They should not be conflated with learning-rate tuning or decay: as we exploit later, the shapes can be designed so that contraction and expansion hold without any reliance on learning-rate decay.

Figure 1(a) illustrates why the shape framing is informative. Repulsion is straightforward: any $f_r > 0$ increases the distance. Attraction is the subtle case, because f_a encodes both attractive and repulsive dynamics: for $f_a \in (-1.0, 0)$ the distance decreases, with a sign flip at $f_a = -0.5$ (maximum attraction, coincident points), while any value below -1.0 actually increases the distance. This dual character of a single “attractive” coefficient is the crux of our second contribution. Although the forces act locally, they collectively sculpt the global structure.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form 1, the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section develops our second contribution: an explanation, grounded in these shapes, of why UMAP needs learning-rate annealing. We focus on UMAP’s default shapes—governed chiefly by a and b —and read off the insights that emerge as we perturb them. The discussion holds generally for $b \leq 1$, where the attraction shape is strictly increasing and hence invertible (see Appendix A for the derivation).

Figure 1(b) plots UMAP’s default attraction shape ($a = 1.58$, $b = 0.89$), which becomes unbounded (approaches ∞) as $\zeta \rightarrow 0$. Exactly as Proposition 4.1 predicts, the contraction-to-expansion transition occurs where λf_a^U crosses -1 as $\zeta \rightarrow 0$. Because the attraction shape is invertible, we can pin down the distance at this transition, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, and interpret it as the *minimum distance for contraction* under attractive updates. In effect, $\zeta > \zeta_{-1}$ contracts and $0 < \zeta < \zeta_{-1}$ expands—directly contradicting the intuition that attractive updates always pull points together. This is the counterintuitive behavior our second contribution explains.

This yields the key takeaway of the contribution: when ζ_{-1} is large, neighboring points oscillate between contraction and expansion and clusters look fuzzy, so achieving sharp boundaries amounts to driving $\zeta_{-1} \rightarrow 0$. Since ζ_{-1} shrinks with λ , this is precisely why UMAP’s schedule must anneal the learning rate to zero (Figs. 1(c,d))—annealing is not a heuristic but a direct consequence of the attraction shape. The repulsion shape (Eq. 12), by contrast, is always positive and satisfies Proposition 4.2: f_r^U approaches 0 as $\zeta \rightarrow \infty$ and ∞ as $\zeta \rightarrow 0$ (Fig. 1(e)).

Comparing attraction shapes across algorithms sharpens the point (Fig. 1(f)): only UMAP and t -SNE must contend with $f_a < -1$ (and hence $\zeta_{-1} > 0$ at $\lambda = 1$). The two cope differently— t -SNE weights its updates by the p_{ij} values, whereas UMAP relies on learning-rate annealing. Neg- t -SNE and PaCMAP, by design, keep f_a within $[-1, 0]$ (satisfying Proposition 4.1 for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$), and PaCMAP’s weighted f_a (< 0.5 throughout) even rules out flips during attraction. The repulsion shapes tell a parallel story: only UMAP incurs large values at small distances (Fig. 1(g); the shape is unbounded, though the values are typically clipped in practice). This cross-method contrast previews the unifying comparison we develop as our fifth contribution; further details appear in Appendix G.

We put this explanation to the test on MNIST by decoupling the learning rates for attraction (λ_a) and repulsion (λ_r) and sweeping each independently (Figs. 1(h,i)). The evidence lines up with the theory: for UMAP, only a small λ_a (< 0.5 , hence $\zeta_{-1} \simeq 0$) yields good embedding (trustworthiness (T) (Venna and Kaski 2001), a measure of structure preservation, Fig. 1(h)) and clustering (silhouette score (SIL) (Rousseeuw 1987), a measure of cluster separation, Fig. 1(i)), whereas the full range of $\lambda_r \in [0, 1]$ works. For NEG- t -SNE and PaCMAP, where $\zeta_{-1} = 0$ by construction, the entire parameter range is effective. This confirms our claim: UMAP needs ζ_{-1} near zero for good embedding and clustering—achieved in practice by annealing—while newer algorithms simply make $\zeta_{-1} = 0$ the default. Additional discussions, results, and illustrations across datasets appear in Section 6 and Appendices B, G and I.

5.1 Improving UMAP’s Consistency under Random Initialization

Our third contribution turns the shape analysis into an actionable fix for a well-known weakness: UMAP’s dependence on careful initialization (Kobak and Linderman 2021; Wang et al. 2021). Standard practice initializes with principal component analysis (PCA) of the data or a spectral decomposition of the high-dimensional graph, which yields consistent mappings despite stochasticity. Under random initialization, however, clusters often fail to form, or form with a different orientation on every run. The shape view explains why: when two high-dimensional nearest neighbors start far apart, the attractive force is too weak to draw them together. This is the near-sightedness phenomenon (Wang et al. 2021), and it is written directly into the attraction shape, where $|f_a^U|$ decays toward zero as distance grows (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| = 0$).

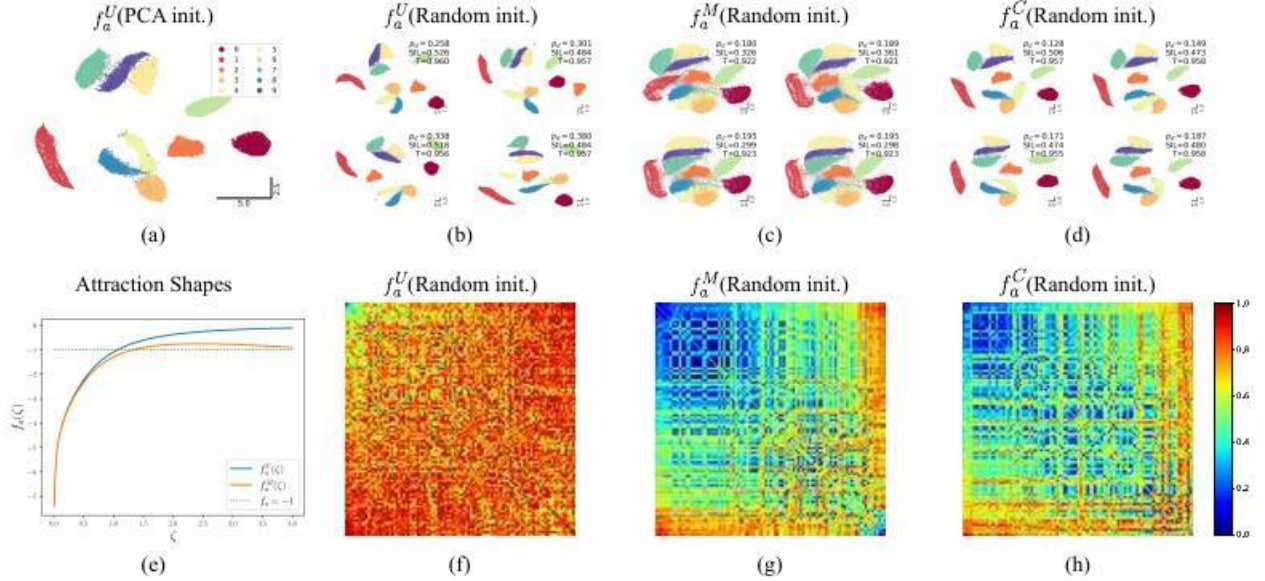


Figure 2: Our third contribution in action: modifying the attraction shape makes MNIST embeddings consistent under random initialization. (a) Reference mapping using PCA. (b-d) The four mappings with lowest Procrustes distance (p_d) from (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP versus modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. Diagonal (i, i) entries are sorted by p_d from (a); off-diagonal (i, j) entries are p_d between the i^{th} and j^{th} mapping. Both the matrices and the (mean $p_d \pm \text{std}$) values make the improvement legible: UMAP’s embeddings are not self-similar, whereas the modified and composite shapes induce scale-invariant structure.

The diagnosis suggests its own remedy: induce “far-sightedness” by boosting attraction at large distances so that faraway neighbors can still be pulled together. To test this hypothesis, we modify UMAP’s attraction shape to strengthen the attractive force:

$$f_a^M = f_a^U - \beta\zeta, \quad (13)$$

where β regulates the strength of the added term (we use $\beta = 0.2$, Fig. 2(e)). Crucially, this shape-level change corresponds to a clean, interpretable regularizer on the attractive loss term (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond pulling in distant pairs, it encourages the point intermixing that aids convergence under random initialization—much like early exaggeration in t -SNE. We chose the simplest linear correction in (13); other choices, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), would serve equally well.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases} \quad (14)$$

The composite shape aims to shed any distortion introduced by f_a^M by reverting to the original UMAP once far-sightedness has done its job. We now quantify how these two shapes—modified and composite—affect DR from random initialization.

We first built a PCA-initialized embedding of MNIST as a reference (Figure 2(a)), then generated embeddings from random (Gaussian) initialization for each shape, repeating the experiment 100 times. To measure consistency, we use Procrustes analysis (Gower 1975), which aligns two point clouds under scaling, translation, rotation, and reflection (details in Appendix D.1); we align each randomly initialized embedding to the PCA reference and report their separation as the Procrustes distance (p_d). The result supports our contribution. For default UMAP, the four lowest- p_d embeddings (Fig. 2(b)) share cluster shapes but not placements, whereas the modified and composite attraction shapes (Figs. 2(c) and (d)) place clusters far more consistently.

To quantify placement more finely, we use the Procrustes matrix, whose diagonal is sorted by p_d from the PCA reference and whose off-diagonal entries give p_d between two randomly initialized mappings (details in Appendix D.1); this is analogous to a similarity matrix (Foote 1999). Default UMAP embeddings are mutually dissimilar (Fig. 2(f)), while

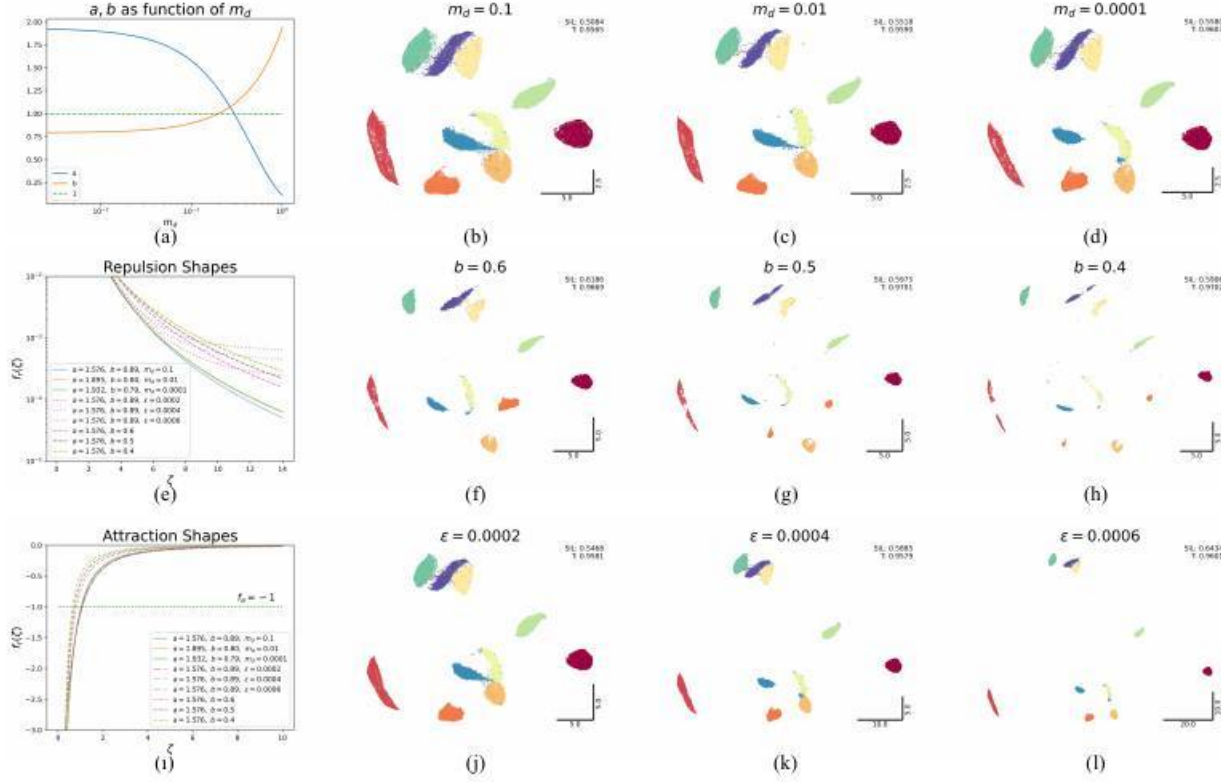


Figure 3: Supporting our fourth contribution: controlling inter-cluster distance on MNIST. (a) Deriving a, b by varying the low-dimensional distance m_d restricts the reachable region. (b-d) Setting m_d to 0.1, 0.01, and 0.0001 barely improves cluster compactness. (e) Repulsion shapes for different parameters. (f-h) Raising repulsion by explicitly varying b yields more compact clusters and even forms clusters absent otherwise. (i) Attraction shapes for the same parameter variations. (j-l) Raising repulsion by adding a small positive value (ϵ) to the repulsion shape widens inter-cluster distance.

the modified (Fig. 2(g)) and composite (Fig. 2(h)) shapes are strongly self-similar. The upshot, and the payoff of this contribution, is that UMAP encodes a unique structure independent of initialization; the difficulty is merely in reaching it, because attraction is too weak at long distances—exactly what our modification repairs.

5.2 Cluster Formation and Compactness

Our fourth contribution uses the shapes to disentangle how clusters form and how compact they become. The parameter practitioners normally reach for is the minimum-distance parameter m_d (through Eq. 2), but varying m_d constrains the accessible values of a and b (Fig. 3(a)), so shrinking m_d often buys no real improvement (Figs. 3(b-d)). The reason is now transparent through the shape lens: m_d barely moves the repulsion shape (Fig. 3(e)). Varying a and b explicitly is the better lever. Decreasing a does raise repulsion, but it suppresses attraction even faster, worsening near-sightedness. Decreasing b gives cleaner control (Figs. 3(e,i)): setting b to 0.6, 0.5, and 0.4 progressively widens inter-cluster distance and splits existing clusters (Figs. 3(f-h)). This splitting reflects the increasingly heavy-tailed kernel as b decreases (heavy-tailed kernels yield smaller, distinct clusters [Van der Maaten and Hinton, 2008, Kobak et al., 2019]; see [Kobak et al., 2019] for how b modulates UMAP’s tail). The approach cleanly separates all ten MNIST labels, but it leaves a puzzle that motivates the next experiment: changing b amplifies repulsion more than m_d does, yet it also reshapes the attraction profile, so the two contributions remain entangled.

To isolate repulsion, we hold attraction fixed and vary repulsion alone—something the shape formulation makes possible for the first time. We modify the repulsion shape by adding a small positive value (ϵ):

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

while keeping a and b fixed, which amounts to a regularizer on the repulsive loss term (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$) and lets us reach stronger repulsion than before (Fig. 3(e)). As ϵ grows, inter-cluster distances grow with it

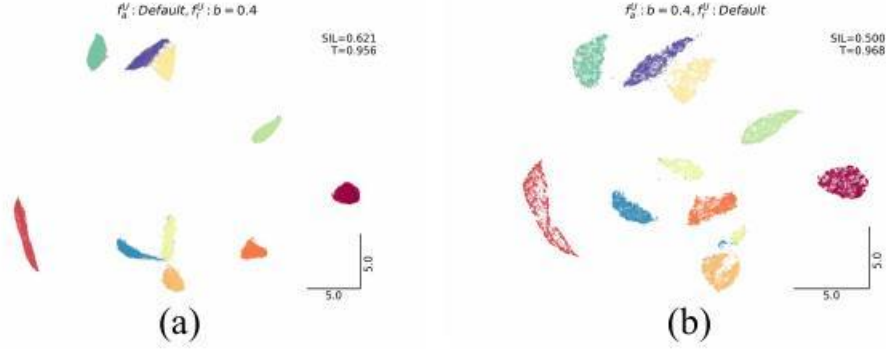


Figure 4: The decisive experiment for our fourth contribution: attraction, not repulsion, drives cluster formation on MNIST. (a) Default attraction with a $b = 0.4$ repulsion shape reproduces default UMAP’s clusters, only more compact. (b) Default repulsion with a $b = 0.4$ attraction shape instead grows new intra-cluster structure and forms new clusters.

(Figs. 3(j-l)), and the clustering behavior mirrors that of varying m_d , giving a loose separation of all labels. The takeaway is pointed: ε leaves the attraction shape untouched and varying m_d traces essentially the same attraction shapes (Fig. 3(i)), which already hints that cluster formation is governed predominantly by attraction.

To resolve the puzzle outright, we vary the attraction shape and the repulsion shape one at a time—holding the other at default—by setting only its b to 0.4 (Fig. 4). With default attraction, the increased repulsion ($b = 0.4$) produces no new structure, only smaller clusters than default UMAP (Fig. 4(a)). With default repulsion, the increased attraction ($b = 0.4$) instead grows additional structure within each cluster and even splits some clusters into smaller ones (Fig. 4(b)). This is the clean result of the contribution: *attraction causes cluster formation, while repulsion controls compactness*—either by shrinking clusters or by widening inter-cluster distance, depending on the repulsion shape. Appendix G.3 shows that LocalMAP embeddings exploit this same attraction-repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

Our fifth contribution puts the shape lens to comparative use, tying three methods together and characterizing their stability. Many DR algorithms fix $a = b = 1$, a choice that keeps gradients bounded and eases optimization. Damrich et al. (2023) recently explored this for UMAP and proposed Neg- t -SNE, which not only stabilizes the gradient but also produces more compact clusters even at a constant learning rate of 1. Separately, Parametric UMAP (Sainburg et al. 2021) began with the UMAP loss but later switched to a numerically stable modified cross-entropy loss (Shi et al. 2023) with a logsigmoid kernel. Viewed through the shapes, these two roads turn out to reach the same destination:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

With $a = 1$ and $b = 1$, the attraction and repulsion shapes of UMAP are given by $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. Even at unity, the shapes reveal the source of UMAP’s fragility: the attraction shape becomes bounded within $[-2, 0]$ with $f_a^U(0) = -2$ (Fig. 1(f)), while the repulsion shape is essentially unchanged (unbounded as $\zeta \rightarrow 0$, Fig. 1(g)). Because $f_a^U < -1$ as $\zeta \rightarrow 0$, Proposition 4.1 and the argument of Section 5 imply that this unity case still demands learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through contrastive embedding (CE) and identified UMAP’s effective kernel as $1/\zeta^2$. Within CE, they introduced NEG- t -SNE by switching the kernel to $1/(1 + \zeta^2)$; recast in UMAP terms this gives the low-dimensional affinity $q_{ij}^N = 1/(2 + \zeta^2)$, and hence the shapes $f_a^N = -2/(2 + \zeta^2)$ and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$ (Figs. 1(f,g)). Here the shape comparison delivers the contribution’s central insight. NEG- t -SNE’s attraction shape is bounded within $[-1, 0]$ and satisfies Proposition 4.1 so any $\lambda \in [0, 1]$ contracts without oscillating; the method is therefore far less sensitive to annealing and stays crisp even at constant $\lambda = 1.0$. Its repulsion shape is likewise bounded within $[0, 1]$ and does not blow up as $\zeta \rightarrow 0$, so fewer points get ejected from clusters under random sampling. Whereas Damrich et al. (2023) attributed this stability solely to bounded (NEG- t -SNE) versus unbounded (UMAP) repulsion, our analysis in Fig. 1 shows that the attraction shape is equally responsible for cluster stability and compactness—a distinction the shape framing makes visible. Further discussion and illustrations appear in Appendix F. Read together with Fig. 4 this points to a broader consequence: attraction and repulsion

shapes from different methods can be mixed and matched—for example, UMAP’s attraction shape with NEG-*t*-SNE’s repulsion shape. We give examples on UMAP, NEG-*t*-SNE, and PaCMAP in Appendix J.

7 Discussion and Conclusion

This work set out to explain, rather than merely observe, how attractive and repulsive forces produce cluster structure. It is well known that attraction pulls similar points together and repulsion pushes dissimilar ones apart, but the precise mechanisms had not been studied. Our unifying move was to treat the attraction and repulsion coefficients as functional *shapes*, and this single reformulation yielded the five contributions we recap below. Although the main text centers on UMAP and NEG-*t*-SNE, the results are general to dimensionality reduction (see Appendix G for other algorithms).

The shape reformulation itself (Contribution 1). Characterizing the shapes through Propositions 4.1 and 4.2 gives exact, per-iteration conditions for contraction and expansion and underpins everything that follows.

Why annealing is necessary (Contribution 2). The shapes exposed a counterintuitive fact: UMAP’s attractive force actually repels (expands distance) at short distances (Fig. 1(b)), pushing neighboring points apart rather than together. UMAP’s learning-rate schedule exists precisely to alleviate this—annealing is a consequence of the attraction shape, not an arbitrary heuristic.

A fix for initialization sensitivity (Contribution 3). Similar points that start far apart feel too little attraction to ever contract. Our formalism supplies a direct remedy—adding a term to the attraction shape to strengthen far-distance attraction—which produced markedly more consistent outputs under random initialization. This gives us confidence that UMAP encodes a unique structure and that numerical tricks merely help reach it faster.

Disentangling cluster formation and compactness (Contribution 4). Analyzing the repulsion shape showed that higher repulsion widens inter-cluster distance, and adding a small positive value to the shape delivered repulsion beyond what varying m_d can achieve. Isolating the two forces revealed the clean division of labor: attraction forms clusters, repulsion controls their compactness.

A unifying comparison (Contribution 5). Finally, comparing NEG-*t*-SNE and UMAP through their shapes showed that, for $a = b = 1$, NEG-*t*-SNE’s bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) satisfy both propositions for any learning rate, explaining its low sensitivity to a fixed $\lambda \in [0, 1]$; varying a and b reintroduces UMAP’s numerical challenges. Along the way we proved that NEG-*t*-SNE and Parametric UMAP are equivalent.

Together, these insights offer concrete new tools for optimizing dimensionality-reduction algorithms. Because DR is closely tied to contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023), we expect the same shape-based approach to benefit representation learning as well. Our broader aim is to make embeddings and their interpretations more principled, consistent, and reliable, and to guide future research.

Software and Data

Every dataset used to support the contributions above is publicly available. The MNIST dataset is available at yann.lecun.com.

Fashion-MNIST is available at github.com.

Single-cell transcriptomes data is available at github.com.

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Appendix

This appendix supplies the derivations, extended experiments, and per-method analyses that back the five contributions of the main text. Appendix A gives the derivations behind UMAP’s attraction and repulsion shapes (Contribution 1). Appendix B revisits the constant-learning-rate story, reinforcing the annealing argument of Contribution 2. Appendix C shows that improved attraction and repulsion also help large language model embeddings, extending Contribution 4 to a new setting. Appendix D.1 formally defines the Procrustes distance and matrix used to quantify Contribution 3, and Appendix E repeats the random-initialization experiment on further datasets. Appendix F deepens the NEG- t -SNE-versus-UMAP comparison of Contribution 5, and Appendix G applies the shape lens to a broader set of DR methods, demonstrating the generality claimed for Contribution 1. Appendix H records, for completeness, how these methods construct the high-dimensional graph. Appendix I reports the detailed λ_a/λ_r sweeps (from Fig. 1) for UMAP, NEG- t -SNE, and PaCMAP across datasets; Appendix J shows the effect of mixing and matching shapes across algorithms; and Appendix K lists implementation details.

A Proofs and Derivations

A.1 Proof of Proposition 4.1

Proof. We subtract Eq. (7) from Eq. (8) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition 4.2

Proof. We subtract Eq. (9) from Eq. (10) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

We use a general form of the low-dimensional affinity function, i.e., $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, to derive the attraction shape. It reduces to UMAP for $\gamma = 1$ and to NEG- t -SNE for $\gamma = 2$. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

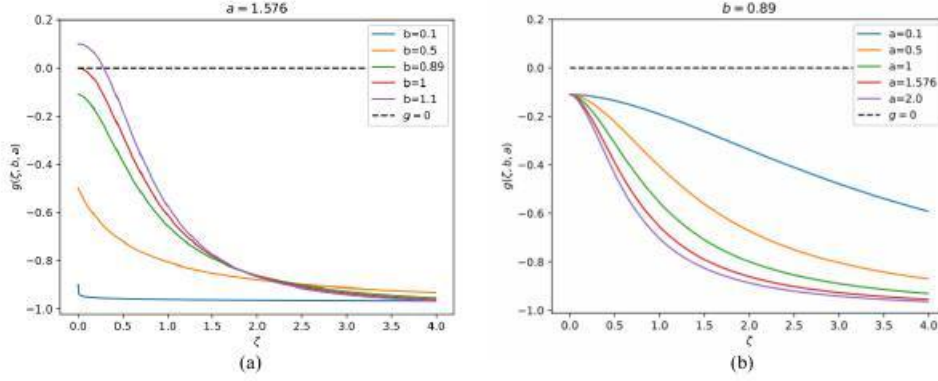


Figure 5: Values of $g(\zeta, b, a)$ for (a) a fixed at 1.576 and (b) b fixed at 0.89.

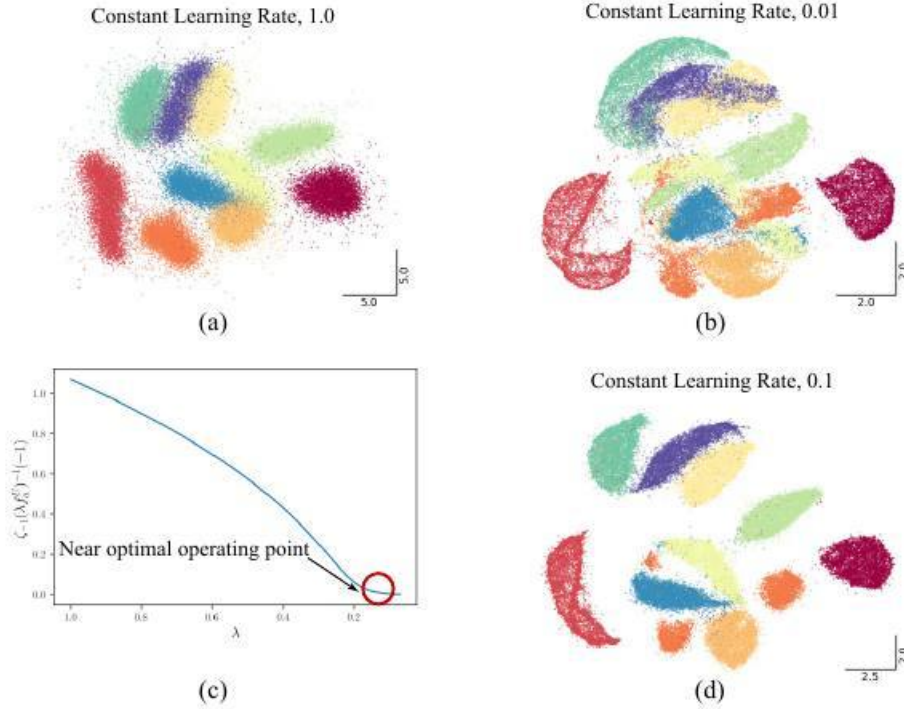


Figure 6: Constant learning rate, viewed through ζ_{-1} (supporting Contribution 2). (a) Too high ($\lambda = 1.0$): the embedding is diffuse because ζ_{-1} is large. (b) Too low ($\lambda = 0.01$): clusters fail to form because attraction and repulsion are too weak. (c) ζ_{-1} falls nonlinearly as the learning rate drops, so the goal is to shrink ζ_{-1} while keeping the forces effective. (d) At the near-optimal constant $\lambda = 0.1$, distinct and compact clusters emerge.

A.4 Condition for strictly increasing f_a^U :

Its behavior with distance can be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

This leads to a strictly increasing condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality is valid as long as $0 < b \leq 1$ (using the derivative and asymptotes of g). Figure 5 shows values of g for different b and a . As b increases above 1, the inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The first term gives the repulsion shape as:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

Generally, $f_r > 0$.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsive term due to Eq. (15) is given by

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases the extra term is a norm-based regularizer. The payoff of working at the level of shapes, however, is interpretability: adding the term directly to the shape makes its role transparent—for attraction it increases far-sightedness, and for repulsion it contributes a constant repulsive coefficient—which is precisely the clarity our reformulation was designed to provide.

B More on UMAP’s Learning Rate

This appendix reinforces Contribution 2 with additional constant-learning-rate evidence. As argued in Sections 5 and 6 UMAP is believed to require learning-rate annealing (Fig 6), and our shape analysis says why. In Section 5 we introduced the minimum distance for contraction (ζ_{-1}) and showed that driving it down via annealing yields compact clusters. In Section 6 we compared UMAP (at $a = b = 1.0$) and Neg- t -SNE and traced Neg- t -SNE’s tolerance of a constant learning rate of 1.0 to its attraction shape lying within $[-1, 0]$, whereas UMAP’s spans $[-2, 0]$. The same logic predicts, and the experiments confirm, that UMAP tolerates a constant learning rate of 0.5 once its attraction shape is confined to $[-1, 0]$ (Figs. 11(f,h)).

Even so, annealing still gives better embeddings across a wide range of a and b , and our framework explains the trade-off precisely. The algorithm’s true goal is to drive ζ_{-1} to zero, which happens only as the learning rate approaches zero; otherwise the embedding stays diffuse (Fig 6(a)). But a learning rate that starts too low makes attraction and repulsion too weak for any clear cluster to form (Fig 6(b)). The ζ_{-1} -versus- λ curve (Fig 6(c)) locates the sweet spot at $\lambda = 0.1$, where ζ_{-1} is small yet attraction and repulsion remain strong. Fixing the learning rate at 0.1 indeed yields compact clusters with clear boundaries (Fig 6(d)).

C Large Language Model Embedding

To show that the cluster-formation insight of Contribution 4 generalizes beyond image data, we apply it to large language model embeddings, where increased attraction and repulsion again pay off. We drew 300,000 PubMedBERT (Gu et al., 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP

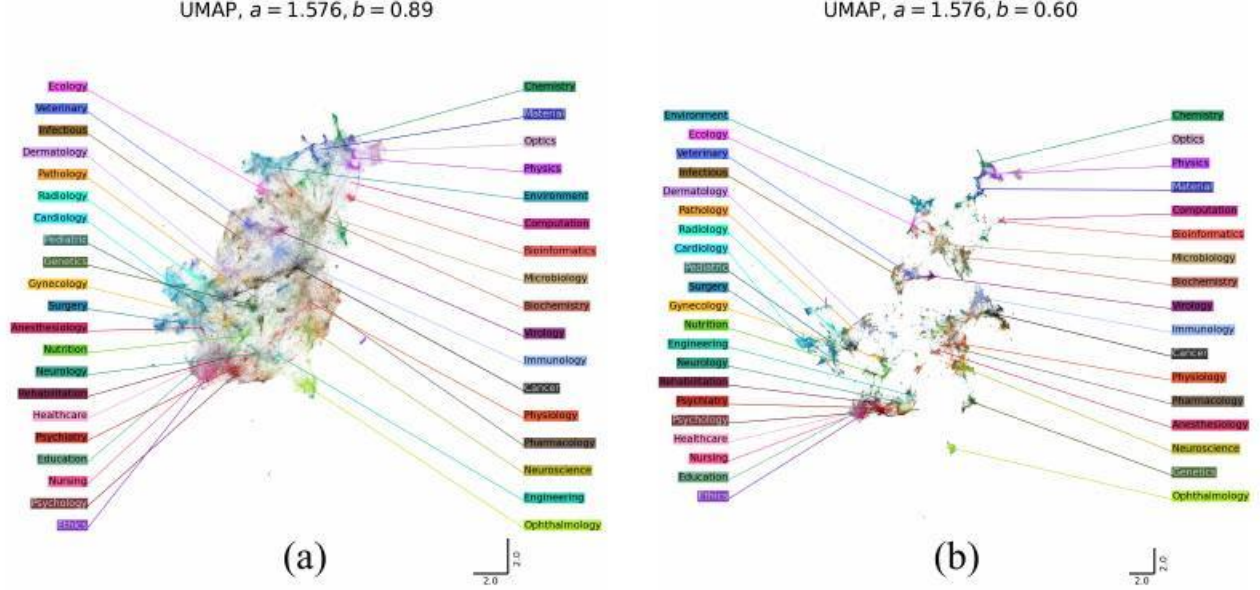


Figure 7: Contribution 4 carried over to LLM embeddings: 300K PubMed abstracts. (a) Default UMAP ($a = 1.576, b = 0.89$) overlaps the labels, whereas (b) UMAP with improved repulsion and attraction ($a = 1.576, b = 0.60$) forms cleaner, non-overlapping clusters. Both mappings are labeled for readability.

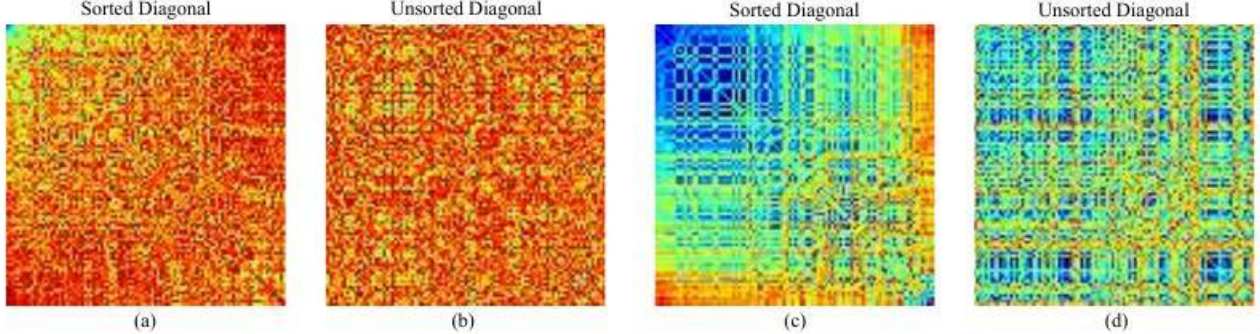


Figure 8: How diagonal sorting affects the Procrustes-matrix visualization used for Contribution 3. (a) The matrix from [2]f) for default UMAP, with the diagonal sorted by p_d of a PCA-initialized sample embedding. (b) The same data with the diagonal left unsorted. (c) The matrix from [2]g) for the modified attraction shape, diagonal sorted as in (a). (d) The same data as (c), unsorted.

yields a crowded map in which the label colors overlap (Fig. 7a); stripped of labels, it would be hard to read, as no explicit clusters appear. Reducing b to 0.6 (following Section 5.2) raises repulsion and, through improved attraction, simultaneously promotes cluster formation, producing a map that is explorable even without labels (Fig. 7b)). The gain is quantifiable: on a 10,000-point hold-out, k -NN accuracy ($k = 10$) rises from 49.2% to 55.48% when b is reduced to 0.6.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

This appendix defines the Procrustes machinery that quantifies Contribution 3. The Procrustes distance (Gower 1975) measures the similarity of two point clouds $\{x\}$ and $\{y\}$ under linear transformations—translation, scaling, and rotation. Operationally, we fix $\{x\}$ and transform $\{y\}$ until the two are maximally aligned. Writing $\{y'\}$ for the aligning

transformation of $\{y\}$, the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

The Procrustes distance is a linear measure that has proven useful in a variety of settings (McInnes et al., 2018; Islam and Fleischer, 2022; Kotlov et al., 2024).

Here, we use the Procrustes distance to measure the consistency of embedding under random initialization. Let $\{x\}_p$ be a reference embedding (using PCA initialization in our experiments), and $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be a set of N embeddings obtained from random initialization. The similarity of the embeddings can be quantified by taking a mean and standard deviation of the strictly lower triangular values of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indexes of X_r can be sorted such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, so that the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal changes nothing numerically, but it makes the embeddings' self-similarity visible—which is exactly what Contribution 3 needs to convey. In Fig. 8(a) the sorted diagonal shows similar embeddings clumping in the upper-left of the matrix. With the modified attraction shape, the set of mutually similar embeddings grows (the larger blue region in Fig. 8(c)), signalling a metastable point in the algorithm. Leaving the diagonal unsorted erases this region and any visible sense of self-similarity (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski, 2001) quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We set $k = 5$, assessing how well each point's five nearest neighbors are preserved. For efficiency we compute trustworthiness on 10,000 randomly sampled indices, using the same indices across embeddings so that the comparison is fair.

D.3 Silhouette Score

The silhouette score (Rousseeuw, 1987) is normally used to assess clustering algorithms; we repurpose it to gauge label separation, i.e., how well an embedding groups the ground-truth labels—the premise being that a good embedding should separate labels as much as possible. For a point y_i in a point cloud $\{y\}$, let a_i be its mean distance to other points sharing its label and b_i the minimal mean distance to points of any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$: close to 1 means y_i sits comfortably within its own label cluster, near 0 marks a boundary point, and close to -1 signals failed label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We computed the silhouette score for the whole embedding (no random sampling) using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

The main-text evidence for Contribution 3 (Section 5.1) used only MNIST. To show the result is not dataset-specific, we repeat the identical experiment on Fashion-MNIST (FMNIST) (Xiao et al., 2017) and single-cell transcriptomes (Macosko et al., 2015) (Figs. 9 and 10). The conclusion holds unchanged: modified and composite attraction shapes—those that strengthen attraction at large distances—substantially improve reconstruction consistency.

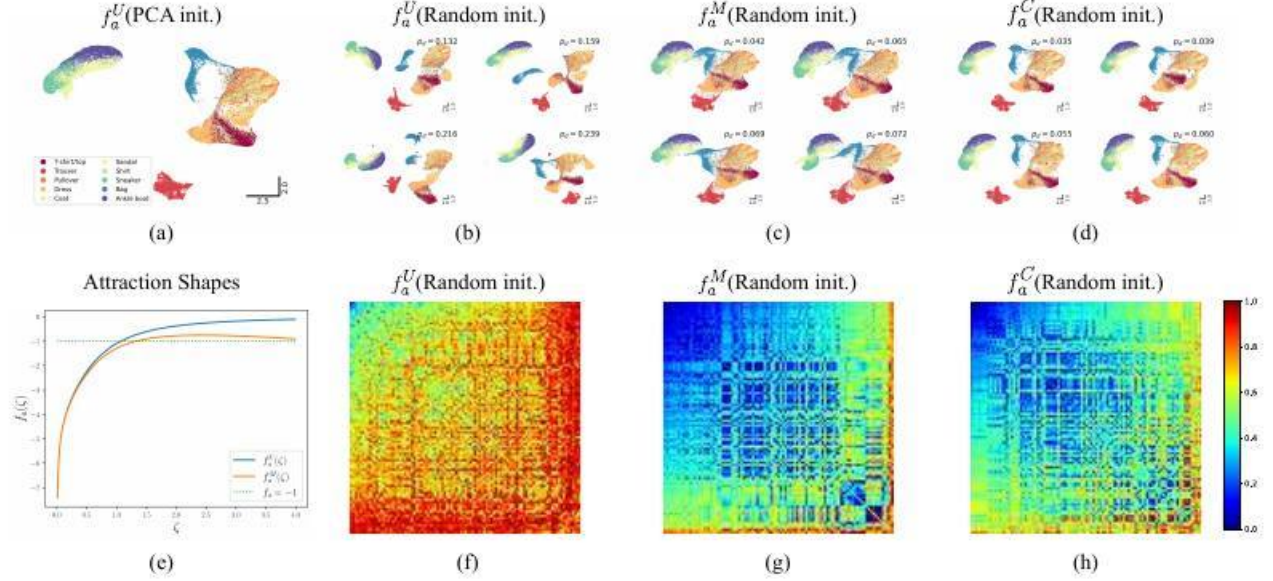


Figure 9: Contribution 3 replicated on FMNIST: random UMAP initialization under different attraction shapes. (a) PCA reference mapping. (b-d) The four lowest- p_d mappings from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default versus modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) shapes.

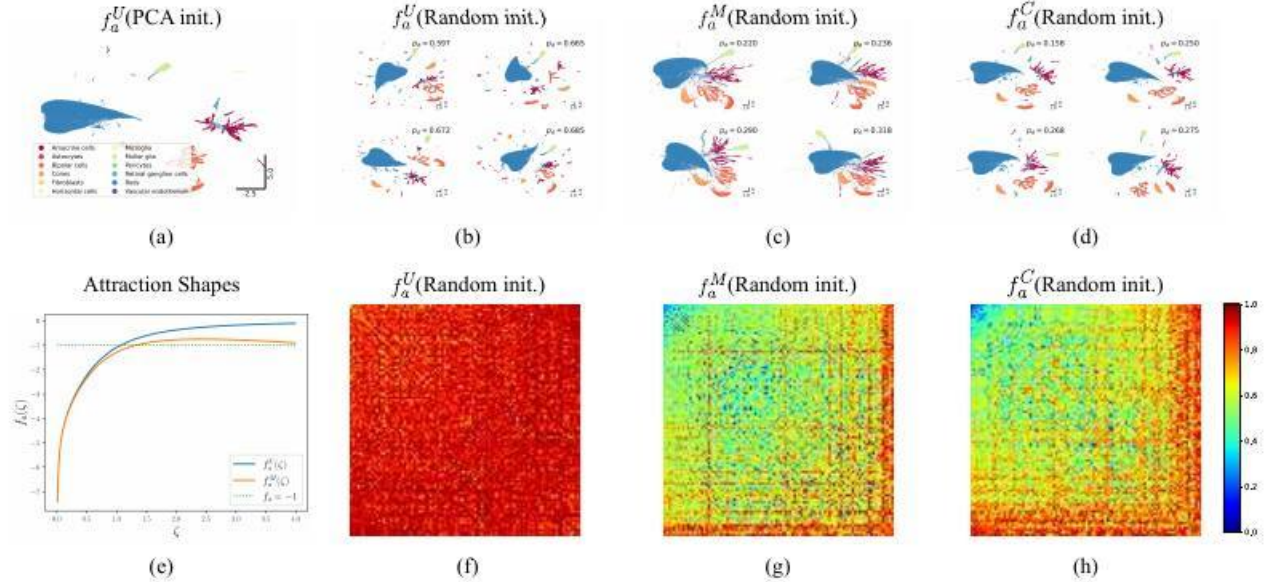


Figure 10: Contribution 3 replicated on single-cell transcriptomes: random UMAP initialization under different attraction shapes. (a) PCA reference mapping. (b-d) The four lowest- p_d mappings from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default versus modified attraction shapes. (f-h) Procrustes matrices from 100 runs of (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) shapes.

F More on NEG-*t*-SNE

F.1 Comparison to UMAP

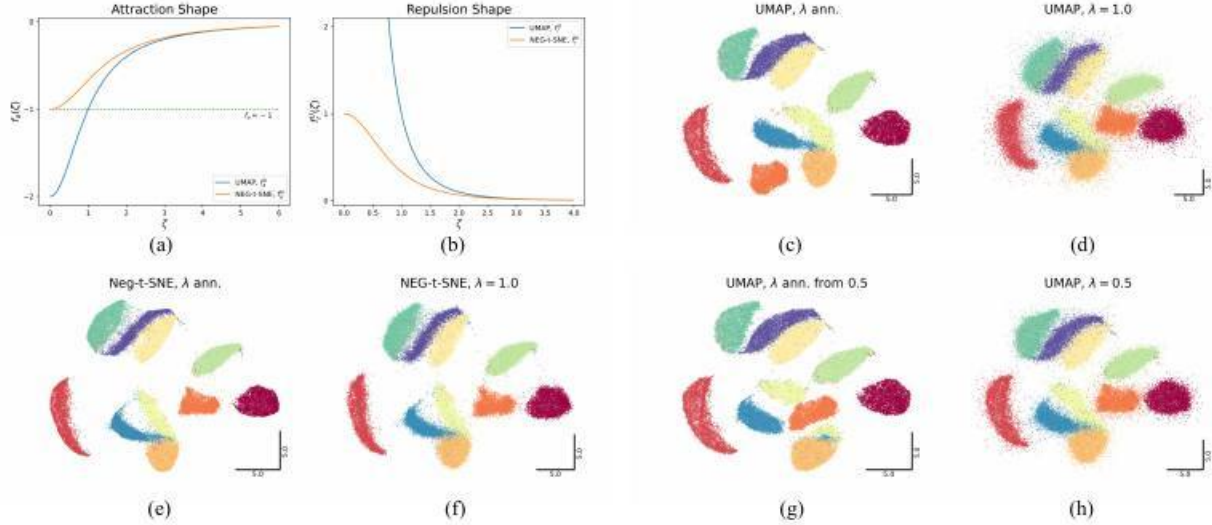


Figure 11: Learning-rate sensitivity of UMAP versus NEG-*t*-SNE on MNIST, the visual backbone of Contribution 5. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG-*t*-SNE. (c,d) UMAP is highly sensitive to λ because $f_a^U < -1$ as the separation ζ shrinks, so clusters turn fuzzy without annealing. (e,f) NEG-*t*-SNE is far less sensitive because $f_a^N \in [-1, 0]$ always, keeping clusters crisp even un-annealed. (g,h) Confining f_a^U to $[-1, 0]$ via $\lambda = 0.5$ likewise reduces UMAP’s sensitivity.

This appendix extends the comparison of Contribution 5. Figure 11 shows the shapes of UMAP and NEG-*t*-SNE together with several MNIST embeddings. Under annealing, the two behave alike (Figs. 11(c,e)); at a constant learning rate of 1, however, UMAP turns fuzzy while NEG-*t*-SNE keeps much cleaner boundaries (Fig. 11(d,f)). Section 6 predicts that confining f_a^U to $[-1, 0]$ should yield less fuzzy clusters at fixed λ —a prediction already visible in Fig. 11(h,i), where UMAP embedded and clustered better once $\lambda_a \lesssim 0.5$. The simplest way to enforce it is to set $\lambda = 0.5$, which satisfies Proposition 4.1 for all ζ . The resulting embeddings (Figs. 11(g,h)) bear out the prediction: clusters resemble NEG-*t*-SNE’s and, at constant $\lambda = 0.5$, are sharper than before (Fig. 11(d,h)). A few stray points remain outside the clusters, a residual effect of UMAP’s repulsion shape that NEG-*t*-SNE avoids.

Next, we can introduce the parameters a and b into NEG-*t*-SNE (essentially the formulation of Parametric UMAP; see Proposition 6.1). The affinity function becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes become $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$, and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes again become unbounded as $\zeta \rightarrow 0$, so NEG-*t*-SNE inherits UMAP’s numerical challenges the moment a and b are allowed to vary—reinforcing the comparative picture of Contribution 5. The one notable difference is that NEG-*t*-SNE’s attraction shape reaches a smaller minimum distance ζ_{-1} : helpful for cluster formation, but it decays to zero faster (more near-sightedness with distance), which can blunt attraction over long ranges.

F.2 Comparison to Parametric UMAP

A further strand of Contribution 5 concerns Parametric UMAP, first trained with the original UMAP objective (Sainburg et al. 2021) and later with a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al. 2023). This change makes Parametric UMAP and Neg-*t*-SNE (Damrich et al. 2023) equivalent (Proposition 6.1), which we prove below.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Both these are NEG-t-SNE with explicit parameters a and b (while in Neg-t-SNE these are set to 1). $\square$

G Alternate Dimensionality Reduction Algorithms

To demonstrate that the shape reformulation of Contribution 1 is not tied to UMAP, this appendix applies it to a broad family of DR algorithms. The alternatives we consider share UMAP’s kernel (with $a = 1$ and $b = 1$ in the low-dimensional weight):

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We proceed from the most involved case to the simplest. We begin with TriMap (Amid and Warmuth 2019): though it relies on triplets rather than pairwise interactions, it serves as a primer for attraction and repulsion that are more intricate than UMAP’s. We then treat Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al. 2021) and its extension with a Local Adjusted Graph (LocalMAP) (Wang et al. 2025), which recast TriMap’s loss for pairwise interactions. Next we tackle t -SNE (Van der Maaten and Hinton 2008), add a short note on SNE (Hinton and Roweis 2002) and its alternate kernel, and close with multidimensional scaling (Borg and Groenen 2007). In every case the shapes make the algorithm’s attraction-repulsion behavior directly comparable to UMAP’s.

G.1 TriMap

TriMap (Amid and Warmuth 2019) optimizes low-dimensional using a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the weight of the triplet (y_i, y_j, y_k) , y_j is in the k -nearest neighbor set of y_i in the high dimension, and y_k is a far-away point. When minimized, we expect that y_i and y_j attract each other, while y_i and y_k repel each other. The update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the distance between nearest neighbors, $\zeta_2 = \|y_i - y_k\|_2$ is the distance from the faraway point, and f_a^T and f_r^T are attraction and repulsion shapes of TriMap, respectively. Unlike UMAP, the attractive and repulsive components are non-separable and the shapes depend on two distance measures (making them 2D). The functional form of the attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

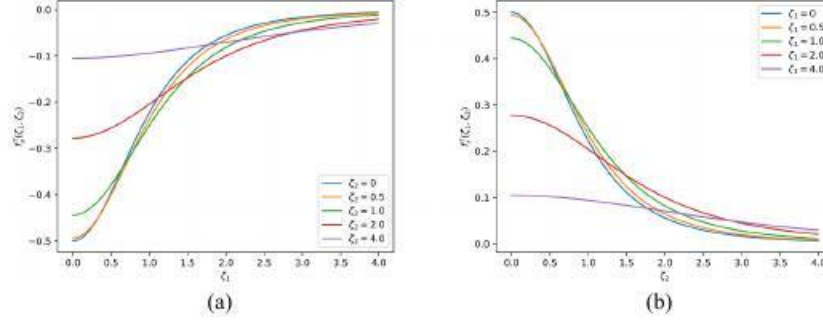


Figure 12: TriMap through the shape lens of Contribution 1: (a) attraction shapes for different ζ_2 and (b) repulsion shapes for different ζ_1 .

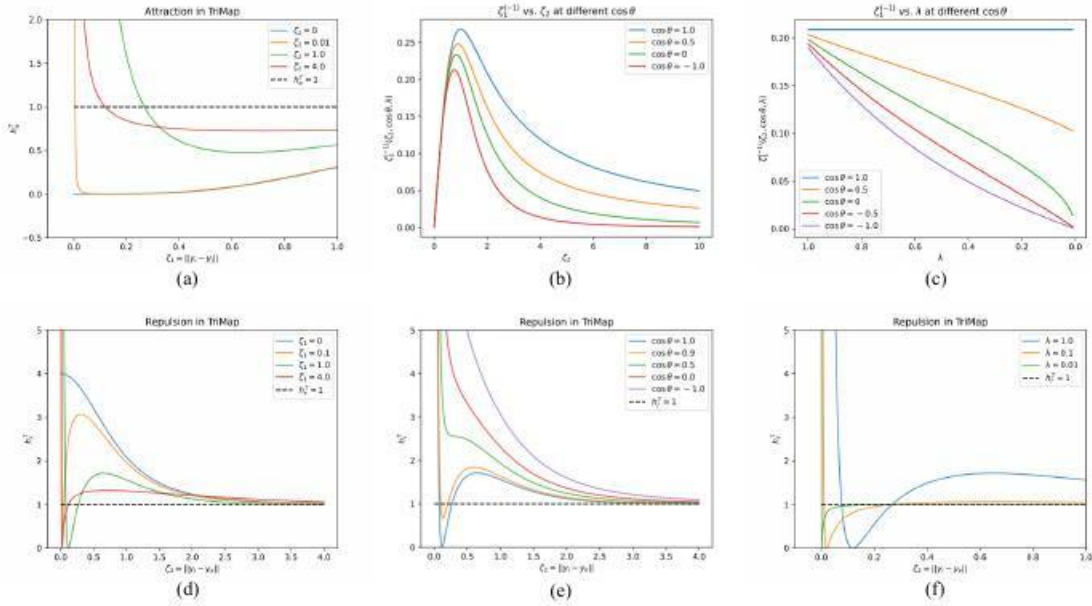


Figure 13: Attraction and repulsion behavior in TriMap, analyzed with the tools of Contribution 1. (a) h_a^T versus ζ_1 for different ζ_2 ; values below the dotted line indicate attraction. As in UMAP, nearest neighbors (y_i, y_j) experience both attraction and repulsion. (b,c) Unlike UMAP, the minimum distance for contraction ($\zeta_1^{(-1)}$) shifts with (b) ζ_2 and (c) λ , with $\cos \theta$ setting its range. (d-f) Repulsion as we vary (d) ζ_1 , (e) $\cos \theta$, and (f) λ ; values above (below) the dotted line indicate repulsion (attraction). Whereas UMAP's repulsive force only repels, TriMap's can do both. Unless labeled otherwise, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap’s attraction and repulsion shapes (Fig. 12) trend like UMAP’s, but with one twist: the repulsion shape bottoms out at -0.5 , so unlike UMAP, attraction alone never flips positions. Because Eqs. (45)-(47) do not decouple attraction from repulsion, Propositions 4.1 and 4.2 do not apply directly; we therefore extend the shape analysis, starting with attraction, to show

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i - y_j)^T (y_i - y_k)}{\|y_i - y_j\|_2 \|y_i - y_k\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$ and λ . In particular, the value of $\zeta_1 = \|y_i - y_j\|_2$, where we want a contraction, is coupled with additional variables. Figure 13(a) shows the attraction behavior for various values of ζ_2 , while $\cos \theta = 1$ and $\lambda = 1$. The values below the dotted line indicate attraction (and thus contraction of distance ζ_1), whereas the values above indicate repulsion (and therefore expansion of distance ζ_1). The value where the dotted line and h_a^T meet gives the minimum distance for contraction ($\zeta_1^{(-1)}$) [analogous to ζ_{-1} of UMAP], which we define as

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and > 0 in most cases (Fig. 13(b,c)). Consequently TriMap behaves much like UMAP and needs learning-rate annealing or an equivalent—echoing Contribution 2—which in practice the authors supply through the delta-bar-delta (Jacobs 1988) method under suitable initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) again depends on ζ_1 , ζ_2 , $\cos \theta$, and λ ; here the quantity of interest is the expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) trace the repulsion behavior as the other quantities vary, with values above (below) the dotted line indicating repulsion (attraction). The striking departure from UMAP, which the shape view exposes, is that TriMap’s repulsion can *contract* rather than expand. Because this anomaly is confined to small distances, appropriate initialization and triplet selection can sidestep it.

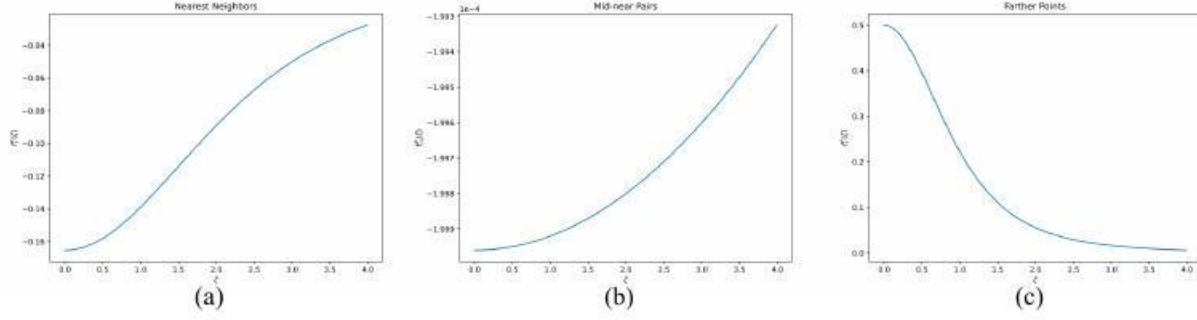


Figure 14: PaCMAP’s attraction and repulsion shapes under the reformulation of Contribution 1. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near points (note the 10^{-4} scale). (c) Repulsion shape for farther pairs. $\lambda = 1$ throughout.

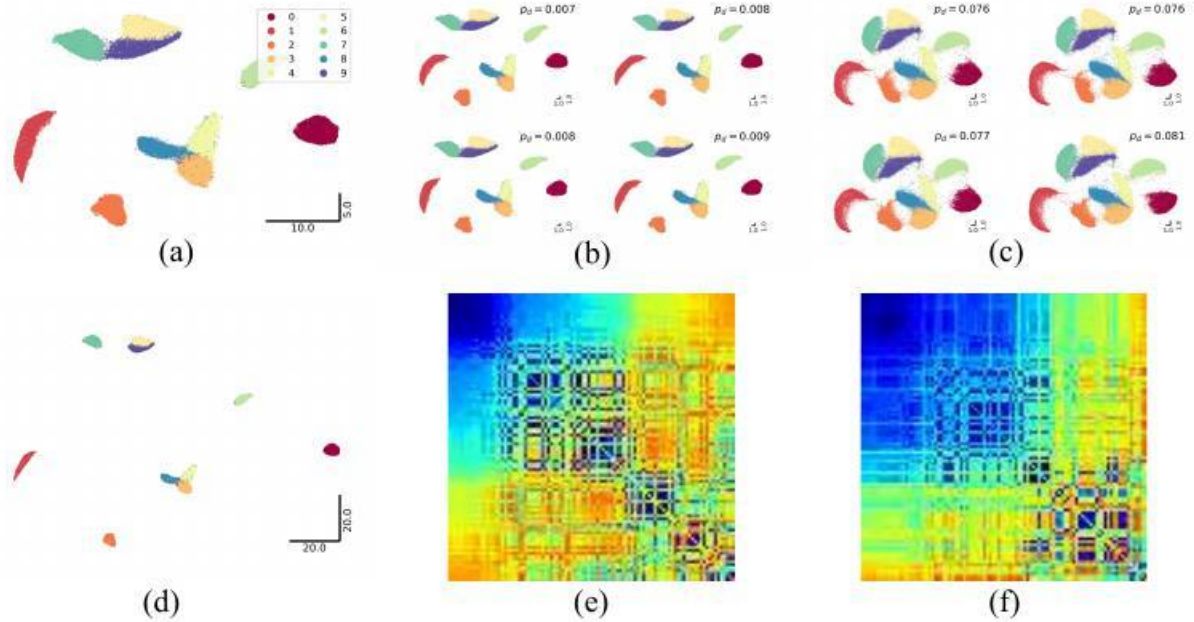


Figure 15: Contributions 3 and 4 carried over to PaCMAP. (a) PaCMAP of MNIST with PCA initialization. (b,c) The four best matches to (a) under random initialization for (b) original PaCMAP and (c) PaCMAP with modified attraction, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP of MNIST with modified repulsion $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) modified-attraction PaCMAP (0.43 ± 0.22). Exactly as in UMAP, stronger far-distance attraction improves consistency, while stronger repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple scales. Its loss is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight the nearest neighbor (NN), mid-near (MN), and further pairs (FP) (details in Appendix H). The first two terms attract and the last repels. The loss is essentially a modified TriMap triplet loss: for the attractive terms it swaps TriMap’s affinity for distant points with constants (1/10 for nearest neighbors, 1/10000 for mid-near pairs, 1 for farther points). Crucially for our analysis, this makes the loss separable into three terms and decouples the update equations—so, unlike TriMap, PaCMAP fits the pairwise shape framework of Contribution 1

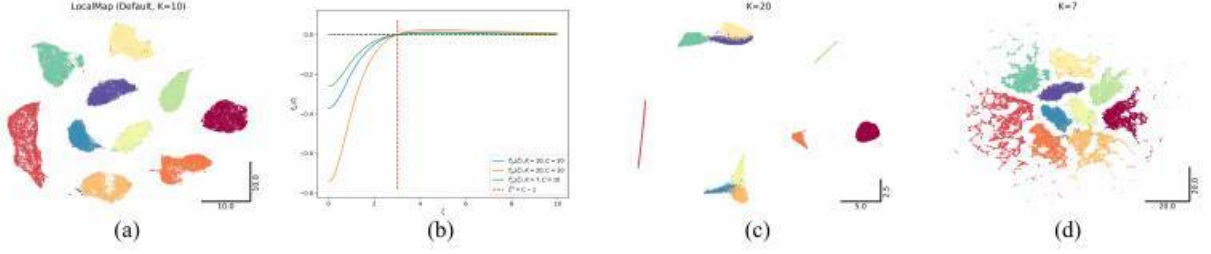


Figure 16: LocalMAP on MNIST, another confirmation of the attraction-repulsion division of labor from Contribution 4. (a) Default embedding. (b) Nearest-neighbor attraction-repulsion shape; values below (above) the dotted line attract (repel), with the transition at $\zeta = \sqrt{C-1}$ (we set $C = 10$, following the LocalMAP implementation). (c) Large K ($= 20$): repulsion dominates and clusters compact. (d) Small K ($= 7$): attraction dominates and clusters break up.

directly. The nearest-neighbor update equations are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

Here f_a^P and f_m^P obey Proposition 4.1 and f_r obeys Proposition 4.2 (Fig. 14), confirming that the propositions of Contribution 1 transfer cleanly to PaCMAP. PaCMAP’s attraction is much weaker than UMAP’s yet suffices across a wide range of learning rates (modulated by Adam (Kingma and Ba, 2015)); with $w_{NB} = 3$ the maximum attraction stays below 0.5, so no flips occur during the attraction update. PaCMAP typically seeds the embedding within a small low-dimensional sphere (on the order of 10^{-3} in (Wang et al., 2021)) and leans on repulsion to pull clusters apart. The net effect is that PaCMAP largely reproduces UMAP’s clustering (especially on MNIST) with improved ordering, and its consistency under random initialization already exceeds UMAP’s (Fig. 15b,e)—consistency that our modified attraction improves further (Fig. 15c,f), just as Contribution 3 predicts. Likewise, adding a small value to the repulsion shape tightens the embedding by widening inter-cluster distance, mirroring Contribution 4.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) refines PaCMAP, and its hallmark—cleanly separating all 10 MNIST clusters (Fig. 16(a))—mirrors the behavior we saw in Figs. 3(g,h,j,k) and Figs. 4(a,b). It thus offers an independent test of Contribution 4’s account of how attraction and repulsion jointly set cluster compactness and connectedness.

LocalMAP first runs PaCMAP and then performs extra optimization on attraction to prise apart certain clusters, minimizing the loss

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term fuses attraction and repulsion into a single coefficient acting on the same pair—attracting in one regime and repelling in the other—which is exactly the kind of dual behavior our shape formulation is built to expose. The second term matches PaCMAP’s, save that LocalMAP resamples further pairs every few iterations. We therefore analyze only the first, nearest-neighbor term, whose update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

By Proposition 4.1 the update contracts as long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$; once $\zeta^2 > C-1$, $f_{ar} > 0$ and Proposition 4.2 makes it expand—so the same two propositions that anchored Contribution 1 govern LocalMAP too. The parameters K and C decide which force dominates. The LocalMAP implementation fixes $C = K = 10$ (Fig. 16(b)), giving stronger attraction than PaCMAP (Fig. 14(a)). As a result, once $\zeta > 3$ the nearest neighbors repel, so pairs bridging two clusters separate, and K —a scaling factor on the forces—tunes how strongly.

Proposition 4.1 also bounds K : requiring $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest attraction f_{ar}^L can supply without flipping pair placements (we avoid such flips so that PaCMAP’s ordering is preserved, since flipping would hinder cluster separation). This reduces (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, i.e., $K \lesssim 13.44$ for $C = 10$, $\lambda = 1$. Above this threshold ($K \gtrsim 13.44$) repulsion dominates and clusters compact, but LocalMAP’s objective breaks down (the inter-cluster bridge survives in Fig. 16(c) for $K = 20$). Below it, attraction dominates (repulsion is too weak) and existing clusters break up (Fig. 16(d) for $K = 7$, echoing Fig. 4(b)).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise distances via the loss

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the weight of the pair. The original implementation of t -SNE considers all the pairs (not just nearest neighbors). This loss function decomposes into attraction and repulsion forces by

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term supplies attraction and the second repulsion. The attractive term is identical to UMAP’s (with $a = b = 1$) and cheap to compute, whereas the repulsive term couples every pair and is therefore costly. Applying the same shape principles we used for UMAP, we can still write down t -SNE’s update equations. Because the original t -SNE forgoes a nearest-neighbor graph, the all-pairs weight $w_{i,j}$ enters those equations directly. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates Eqs. (72)-(73), f_a^{t-SNE} obeys Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$) and defines the minimum distance for contraction ζ_{-1} . The repulsive update, coupled across all pairs, has no such simple relation to the repulsion shape; instead, reusing the machinery developed for TriMap, we obtain:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because t -SNE's repulsion condition (Eq. 78) has the same form as TriMap's, its repulsion behaves the same way— t -SNE's repulsive forces can attract as well as repel.

Since t -SNE scales its forces (by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion), both forces are typically weaker than UMAP's, so t -SNE relies on large learning rates (e.g., $\approx 10^3$ in Open- t -SNE (Poličar et al. 2024)). It also needs an 'early exaggeration' step that multiplies the attractive forces by a constant for the first few iterations. This does exactly what our far-sightedness analysis (Contribution 3) prescribes: points that belong together but sit far apart are drawn in, and points that are too close ($\zeta_{i,j} \approx \zeta_{-1}$) yet should separate are pushed out. Early exaggeration is thus indispensable for finding a consistent t -SNE embedding, especially under random initialization—a striking parallel to our own attraction modification. (Applied throughout optimization, the same trick pushes t -SNE embeddings toward UMAP's (Böhm et al. 2022).)

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis 2002), among the oldest algorithms in this class, uses the same loss as t -SNE (Eq. 71) but with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. Under the shape lens this yields the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 (replacing the t -SNE symbols with their SNE counterparts). The shape view immediately explains SNE's weakness: a constant attraction shape is ill-posed and leans entirely on λ and $w_{i,j}$ for contraction, so clusters overlap even for few samples (the crowding problem (Van der Maaten and Hinton 2008))—the very defect t -SNE and its successors cure by trading the Gaussian kernel for a heavy-tailed one, Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen 2007) usually avoids gradient methods—they often fail to converge to good mappings—and uses stress majorization instead, though a few works do pursue gradients (Kruskal 1964; Zheng et al. 2018). We treat the gradient case briefly, since it too fits our shape framework. In particular, Zheng et al. (2018) give a successful gradient-descent MDS for graph drawing. Starting from the cost function (we drop $w_{i,j}$ without loss of generality, folding it into the learning rate):

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

Each pair contributes a single term, with no explicit attractive and repulsive split—so that lone term must furnish both attraction and repulsion, precisely the dual-role situation our shapes handle well. Because the loss separates into independent (i, j) components, we can analyze one pair at a time. Its derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

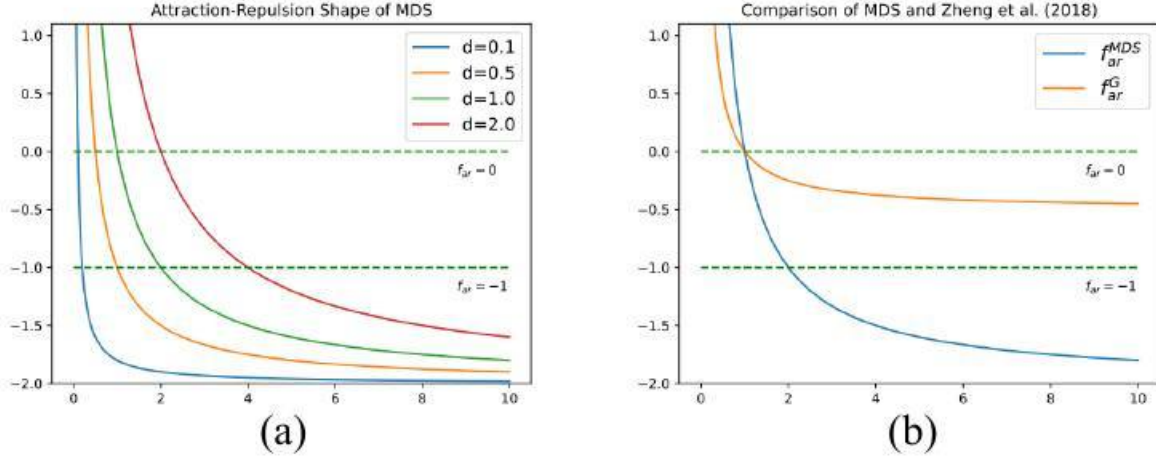


Figure 17: MDS viewed as a single attraction-repulsion shape (Contribution 1). (a) The attraction-repulsion characteristic f_{ar} of MDS. (b) MDS versus the method of (Zheng et al., 2018) for $d = 1.0$.

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape attracts when $\zeta > d$, and in that range is bounded within $[-2, 0]$ (Fig. 17(a)). But the shape reveals why convergence is hard: only values in $[-1, 0]$ contract, the rest expand, and values below -2 flip points and expand. To sidestep this, (Zheng et al., 2018) adopt an ad-hoc, force-directed-graph-inspired form,

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

which is bounded within $[-0.5, 0]$ and hence works for an effective learning rate $\lambda \leq 1$. For $\zeta < d$, $f_{ar}^{MDS}(> 0)$ instead repels. Overall this interaction works best when initialization is already near the target: repeated optimization of a pair oscillates around $\zeta = d$ (where $f_{ar}^G = 0$), giving a notion analogous to UMAP’s ζ_{-1} . No learning rate drives this distance to zero—indeed, that oscillation *is* MDS’s objective—which, in the language of Contribution 2, is exactly why MDS clusters look fuzzier than UMAP’s or PaCMAP’s (for illustrations see (Lambert et al., 2022), (de Bodt et al., 2025), and (Kury et al., 2025)). Note that (Lambert et al., 2022) recast the MDS loss as a quartet stress (over four samples) with a relative distance (divided by the sum of the six quartet distances), enabling global regularization at lower cost than majorization.

H Construction of the High-Dimensional Graph

For completeness, we record here how these methods build the high-dimensional graph—a step that precedes, and is orthogonal to, the shape-based optimization analysis that carries our contributions. Stochastic Neighbor Embedding (SNE) (Hinton and Roweis, 2002), which underpins the modern algorithms, constructs a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$ via

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match a user-defined value *perplexity*, P , defined as

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP, by contrast, builds its high-dimensional graph from the k -nearest-neighbor (k -NN) algorithm via

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k -nearest neighbors of the point x_i and σ_i is a scaling parameter such that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses just the k -NN graph for the affinities (all equal to 1) with a self-tuning distance measure (Zelnik-Manor and Perona 2004),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance from x_i to its fourth through sixth Euclidean nearest neighbors. Despite their differing definitions, this σ_i plays the same role as those in Eqs. (87) and (90).

Whatever these graph-construction choices, our analysis shows that the low-dimensional embedding is driven chiefly by the optimization—and hence by the attraction and repulsion shapes at the heart of this paper.

I Detailed Results for Varying λ_a and λ_b

This section provides the full evidence behind the λ_a/λ_r sweeps of Fig. 11(h,i), which support Contributions 2 and 5. We obtained the results by swapping UMAP’s attraction and repulsion shapes for those of the other methods. Figure 18 reproduces the main-text summary. Figures 19-22 give, for each method, the individual embeddings across choices of λ_a and λ_r , together with the initialization, a reference embedding under annealed λ , and the cases $\lambda_a = 0$ or $\lambda_r = 0$. For PaCMAP, we additionally include a mid-near-pairs reference. Figures 23-27 and 28-32 repeat everything on FMNIST and single-cell transcriptomes, respectively.

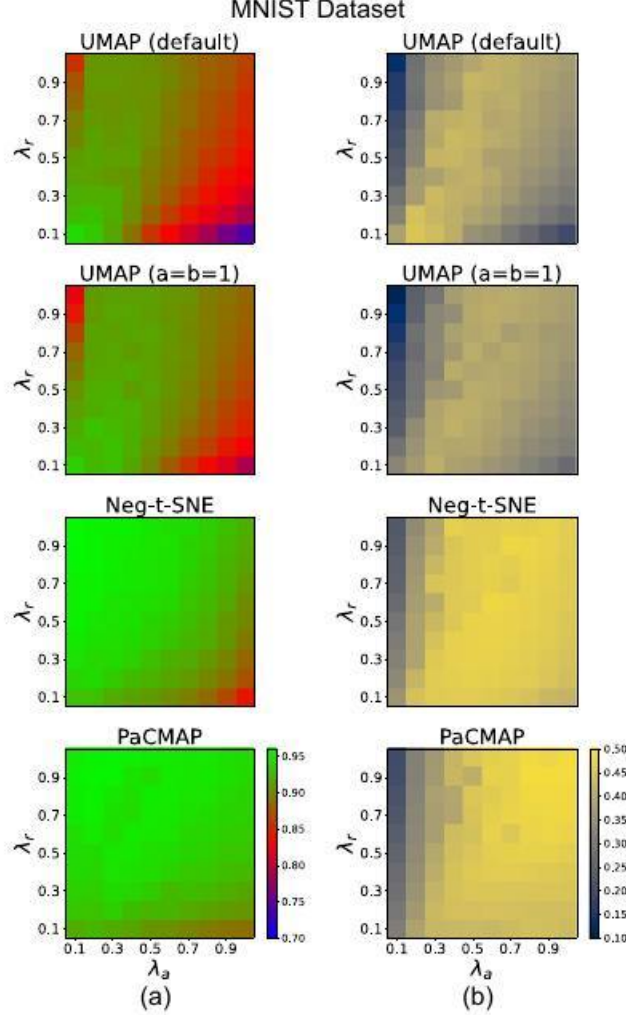


Figure 18: Method comparison on MNIST underlying Contributions 2 and 5: (a) trustworthiness and (b) silhouette score across methods.

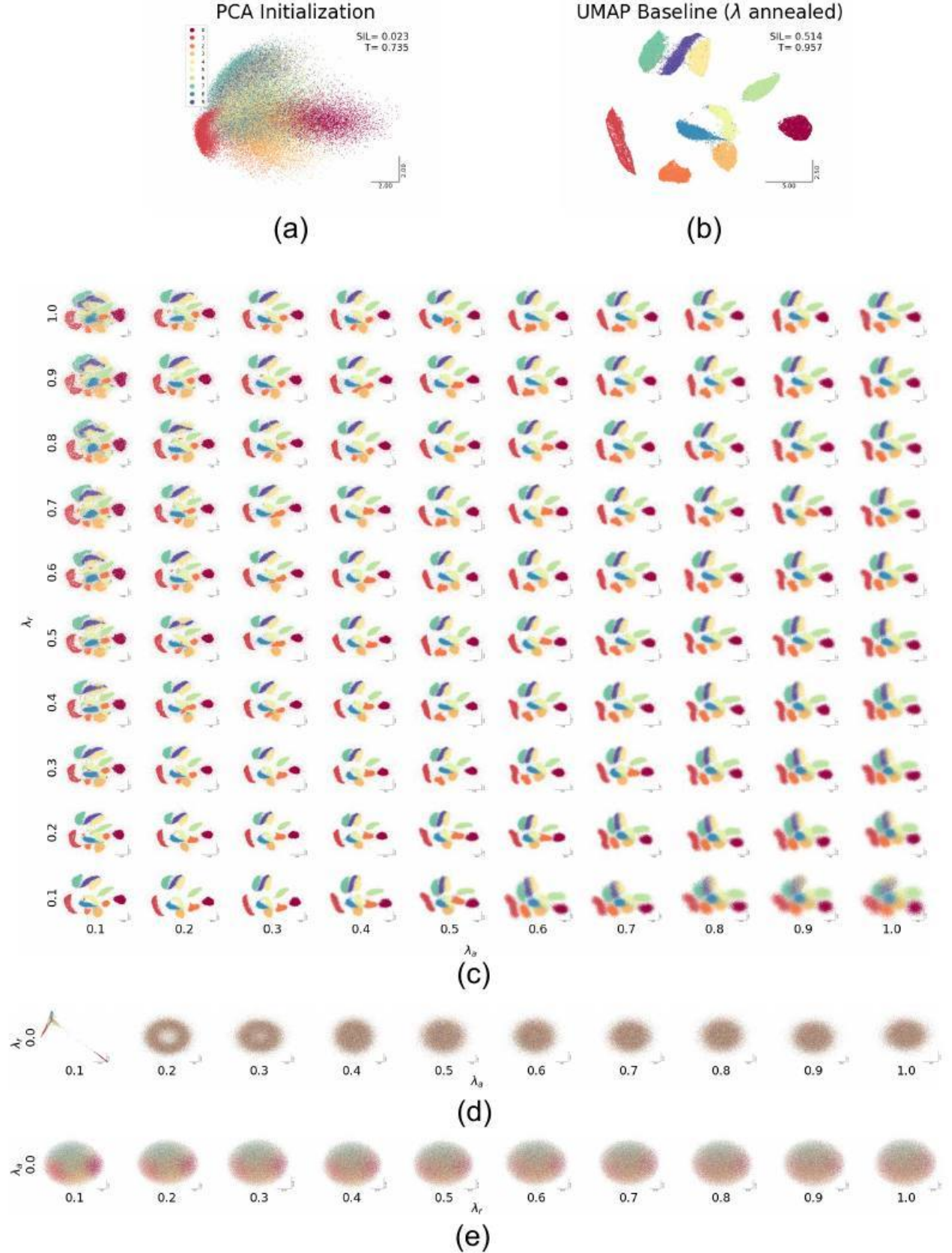


Figure 19: Varying λ_a and λ_r for the MNIST dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

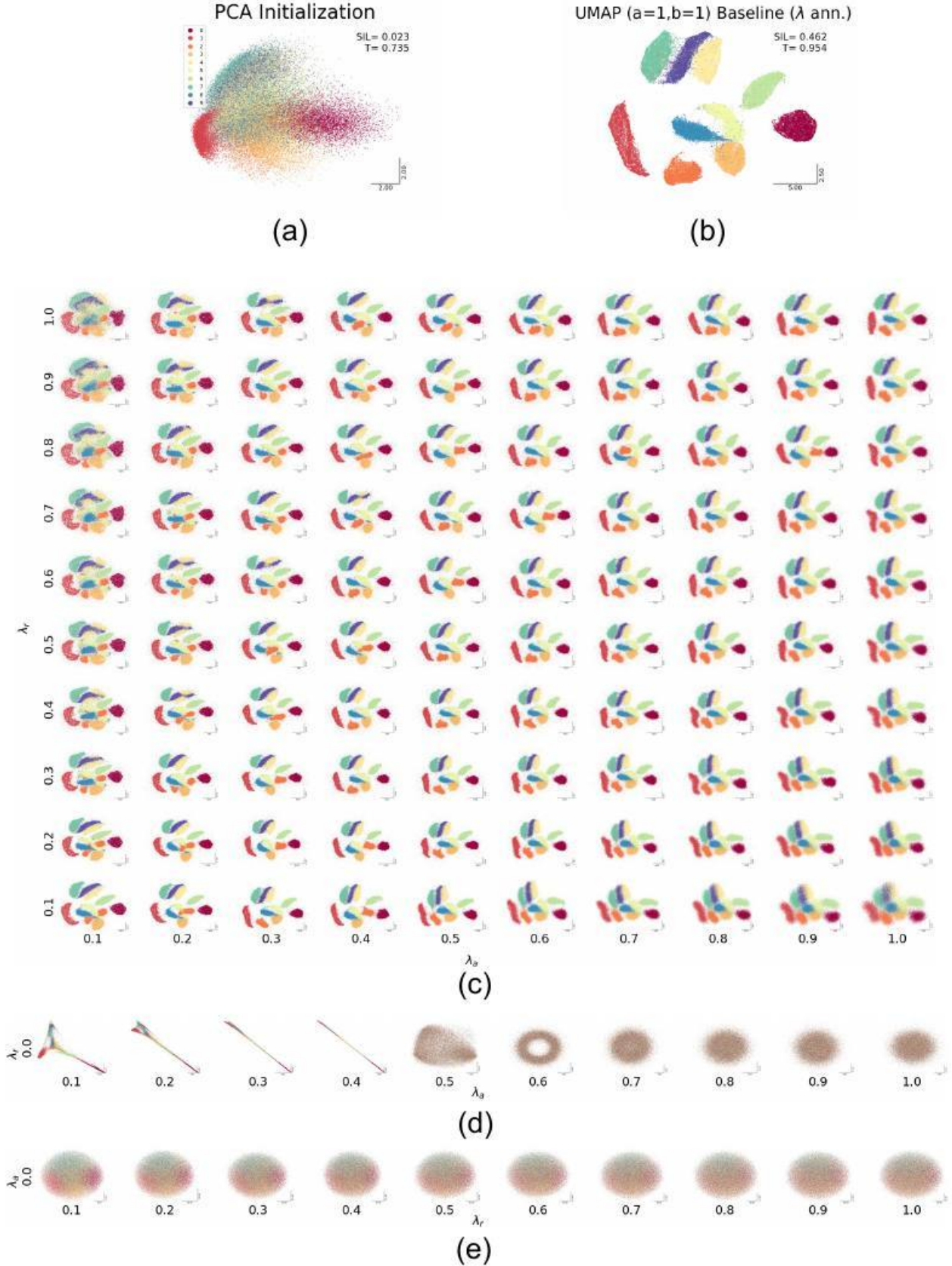


Figure 20: Varying λ_a and λ_r for the MNIST dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

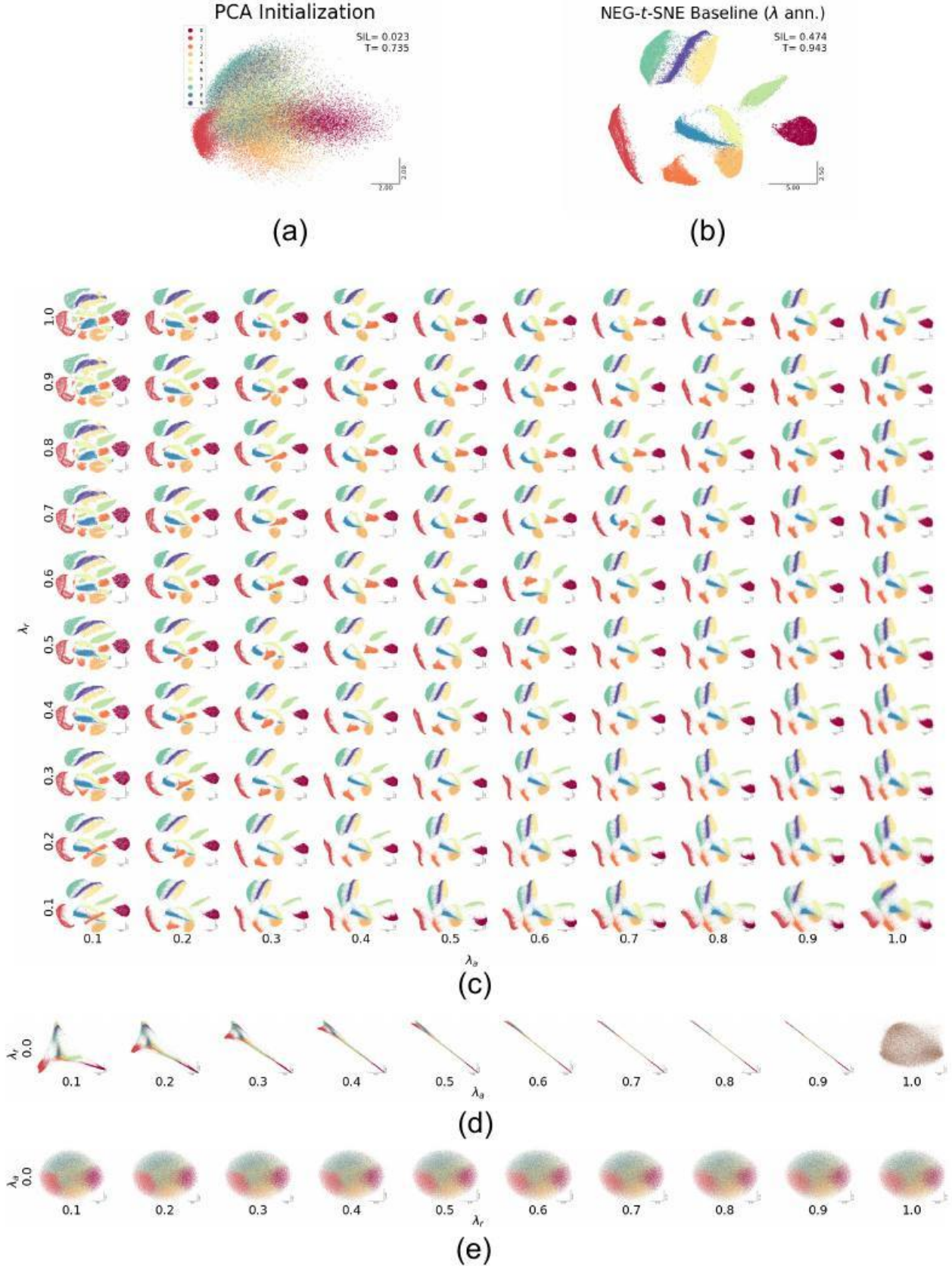


Figure 21: Varying λ_a and λ_r for the MNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

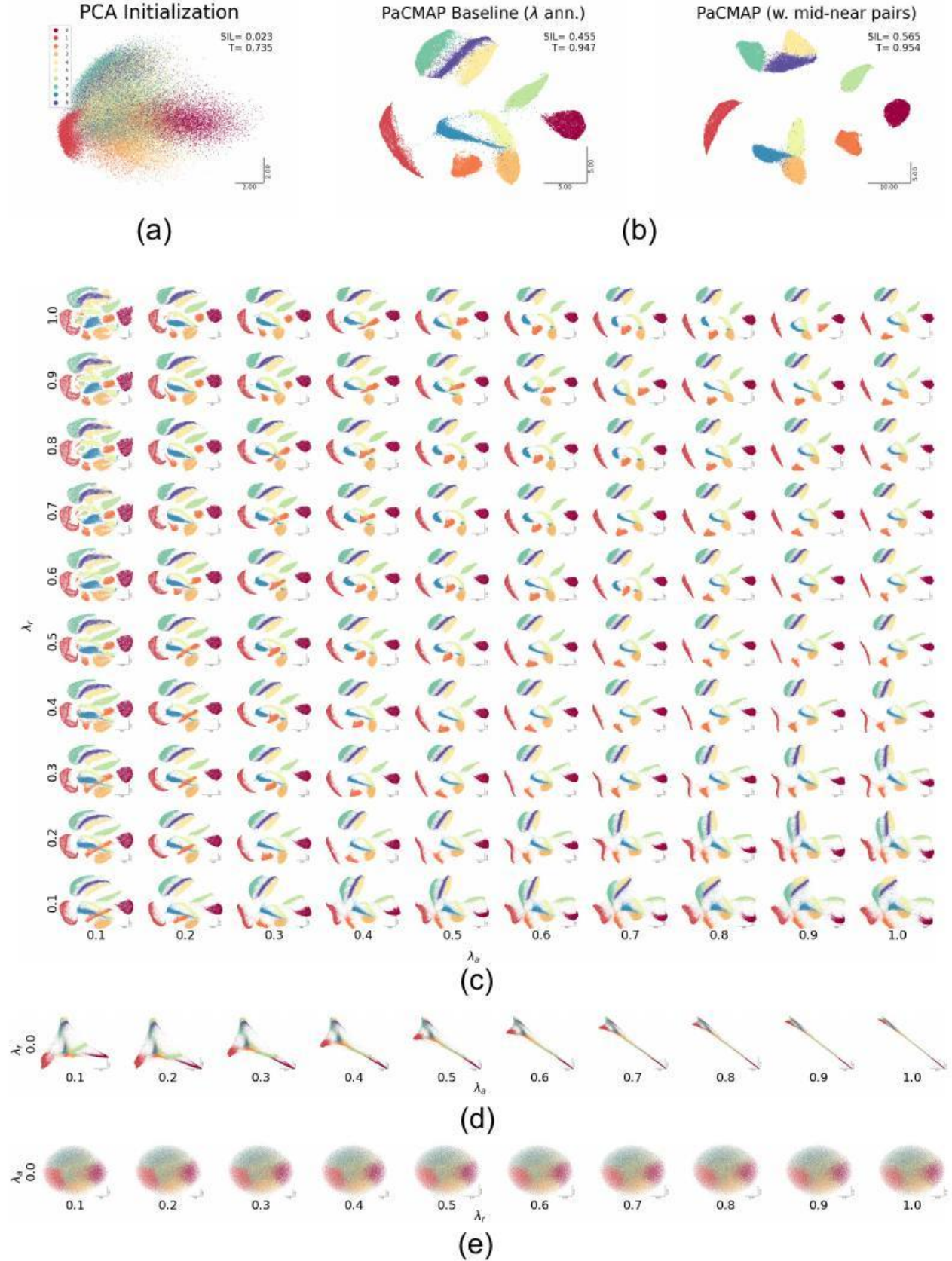


Figure 22: Varying λ_a and λ_r for the MNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

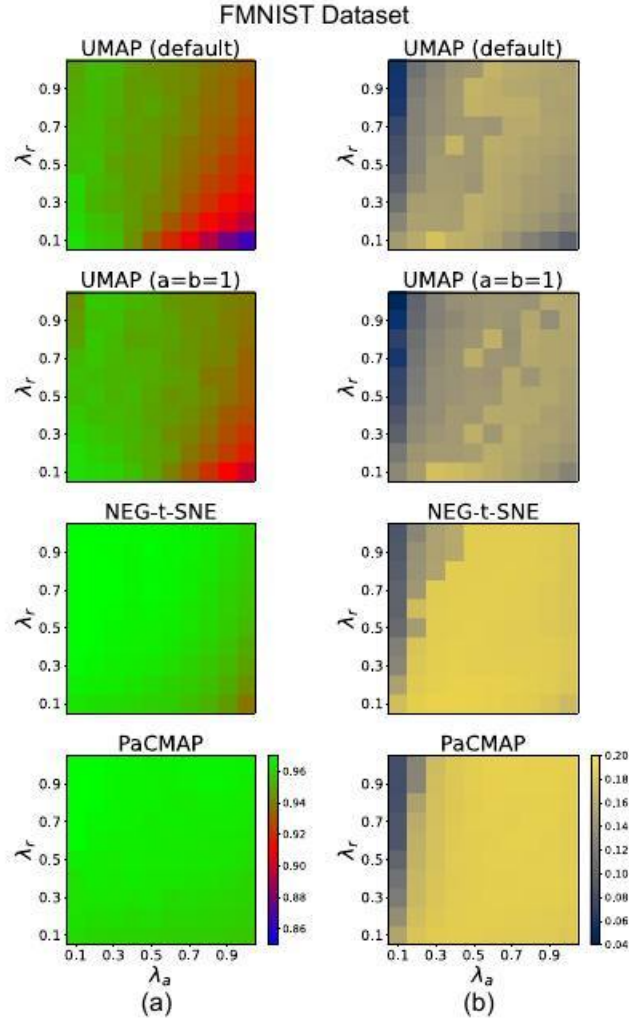


Figure 23: Method comparison on FMNIST underlying Contributions 2 and 5: (a) trustworthiness and (b) silhouette score across methods.

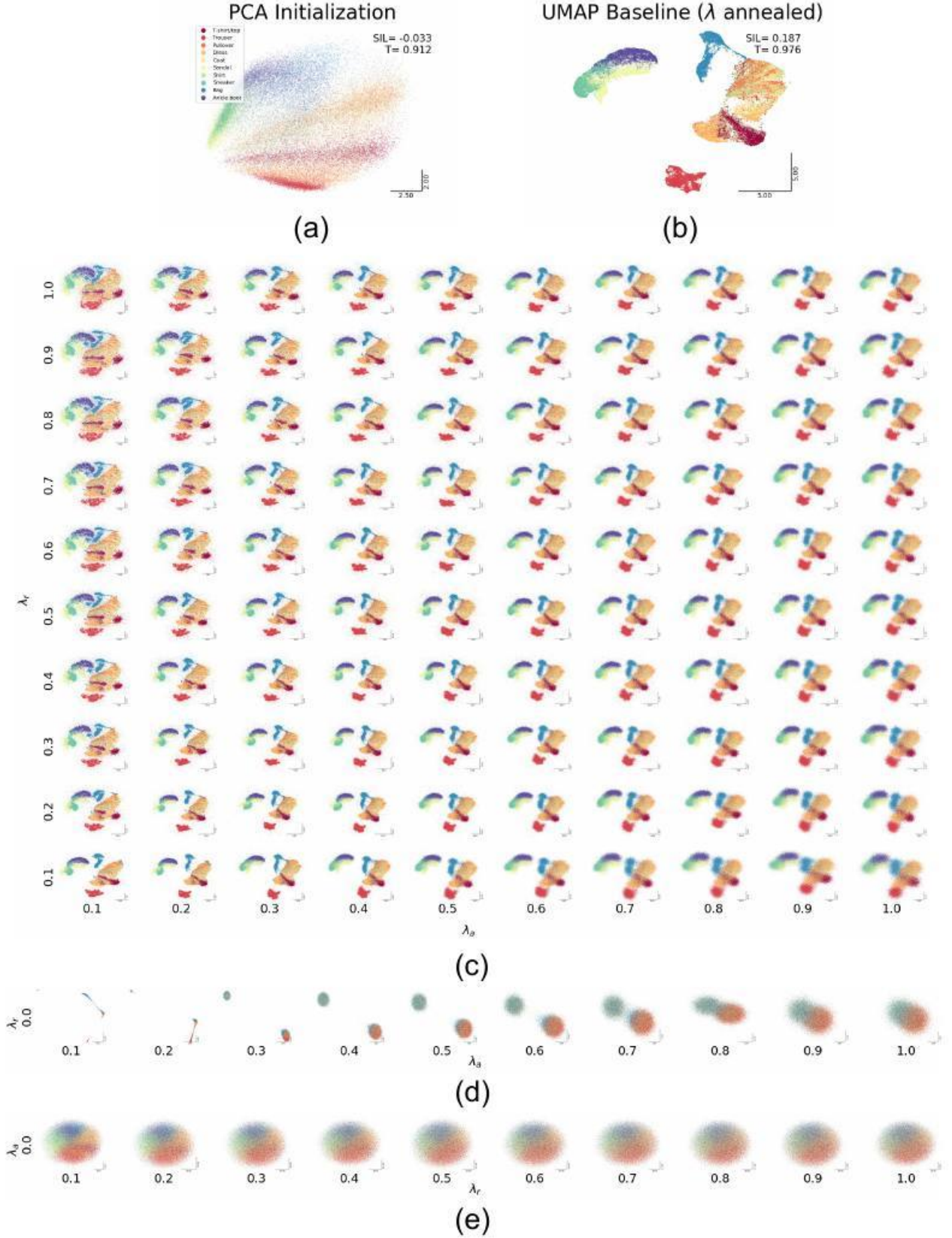


Figure 24: Varying λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

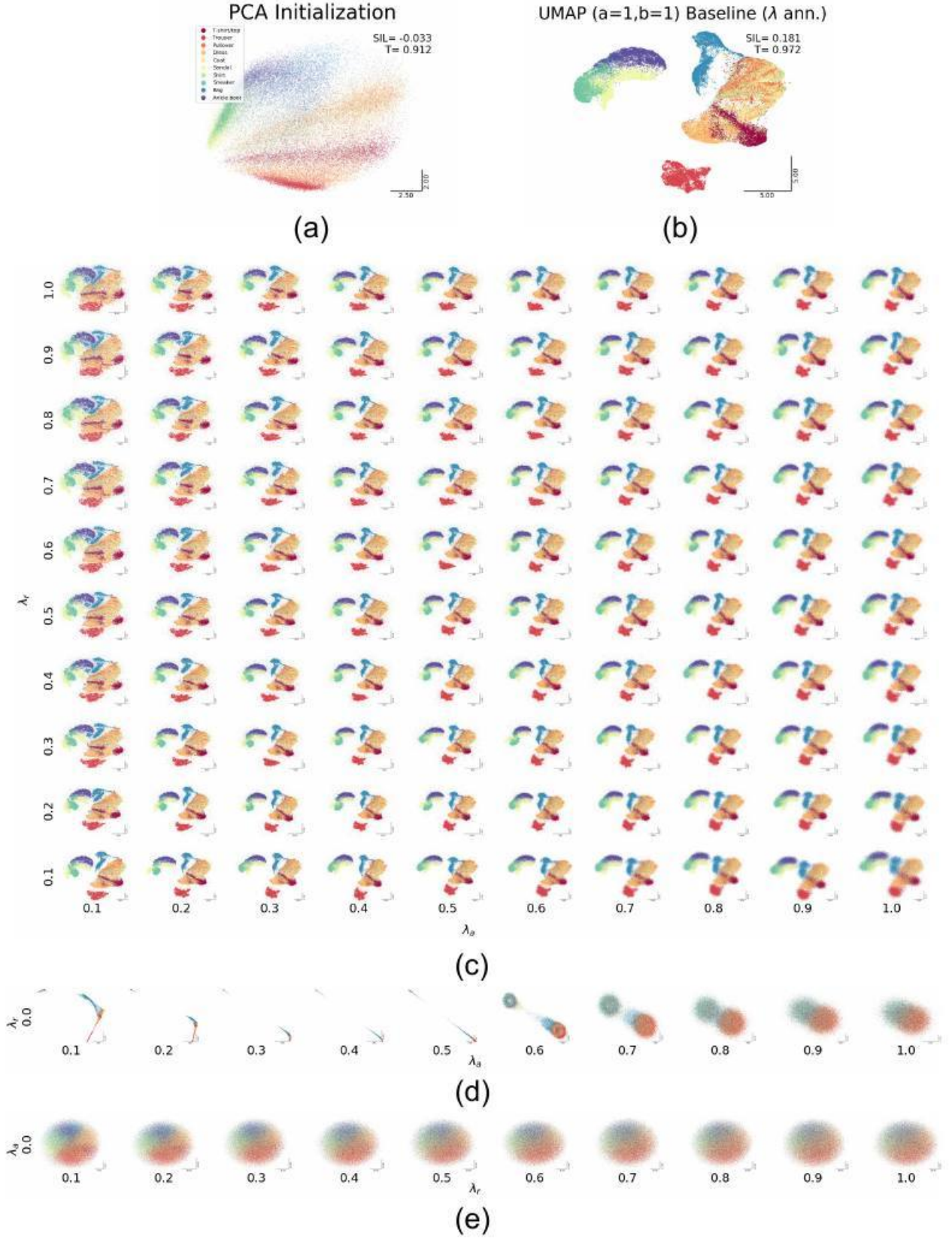


Figure 25: Varying λ_a and λ_r for the FMNIST dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

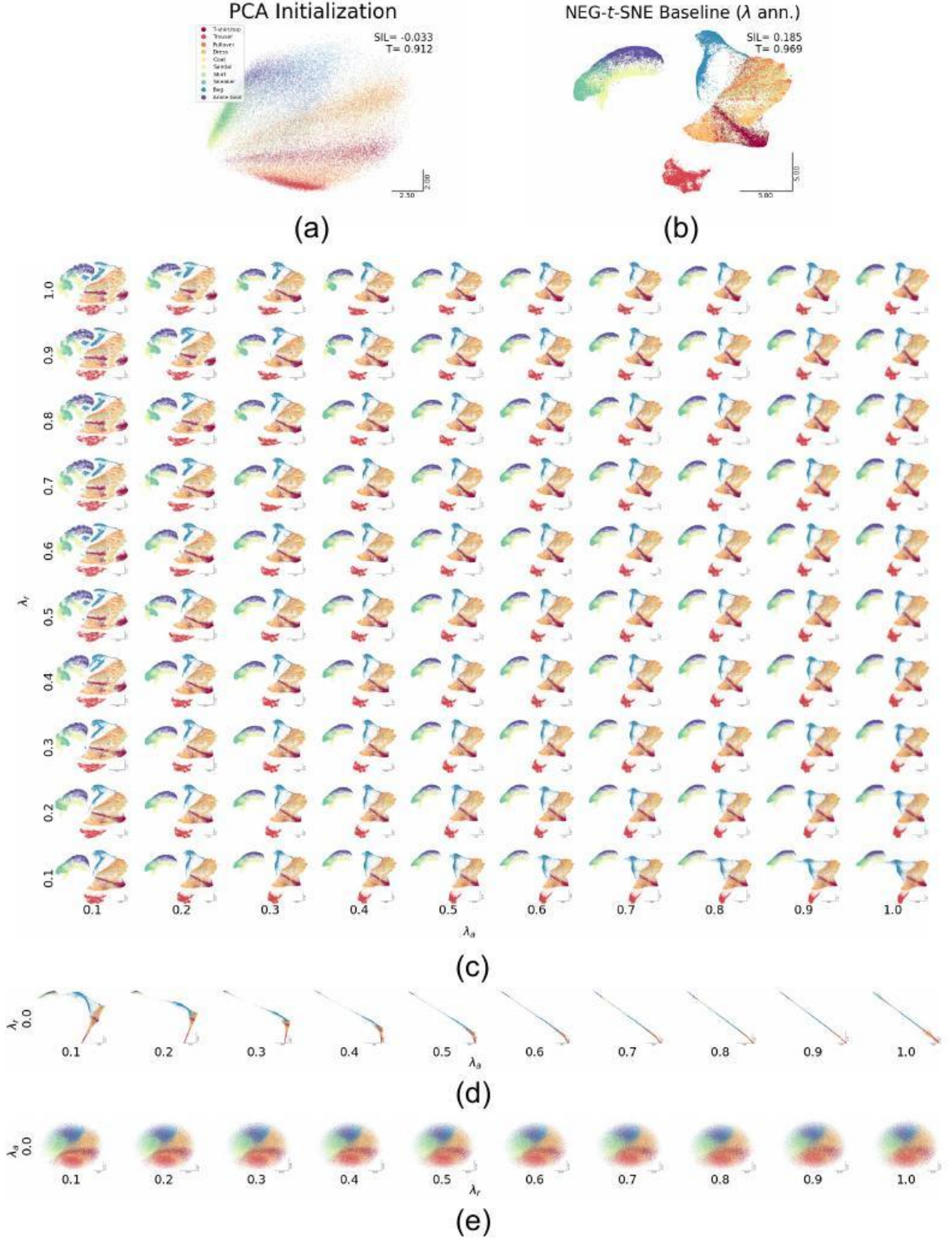


Figure 26: Varying λ_a and λ_r for the FMNIST dataset using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

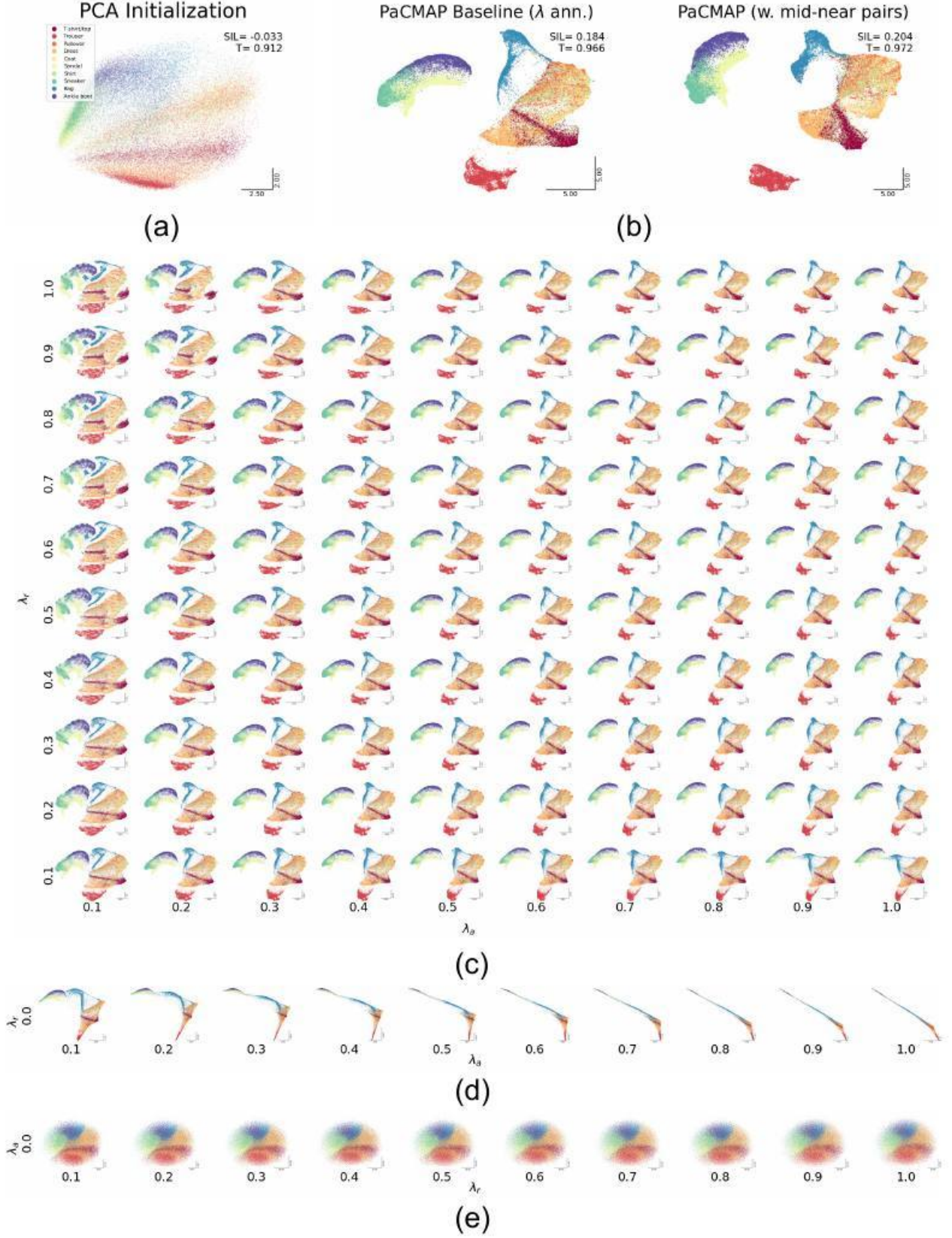


Figure 27: Varying λ_a and λ_r for the FMNIST dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

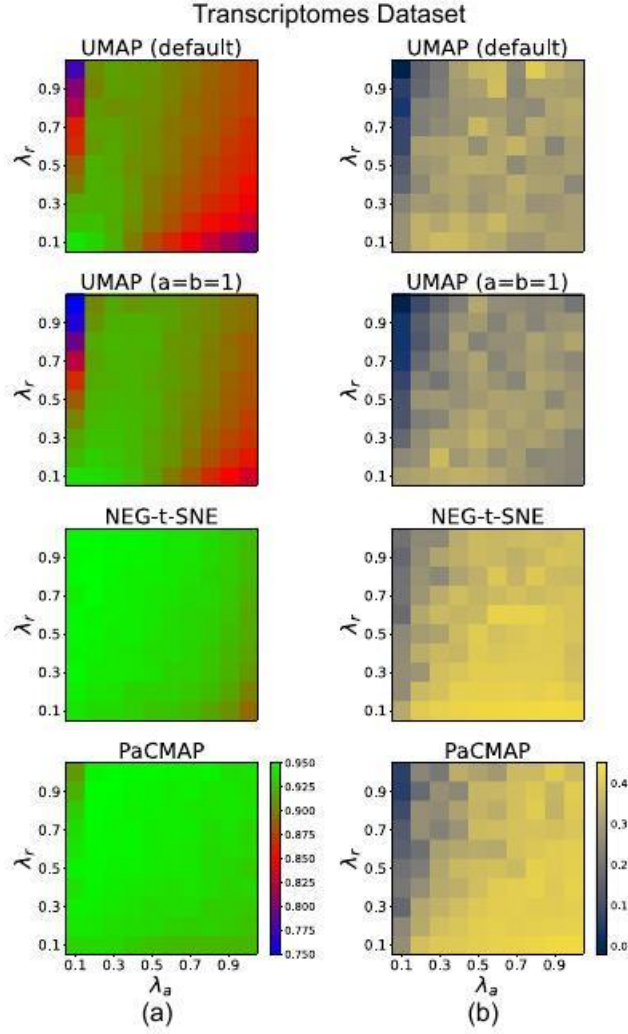


Figure 28: Method comparison on single-cell transcriptomes underlying Contributions 2 and 5: (a) trustworthiness and (b) silhouette score across methods.

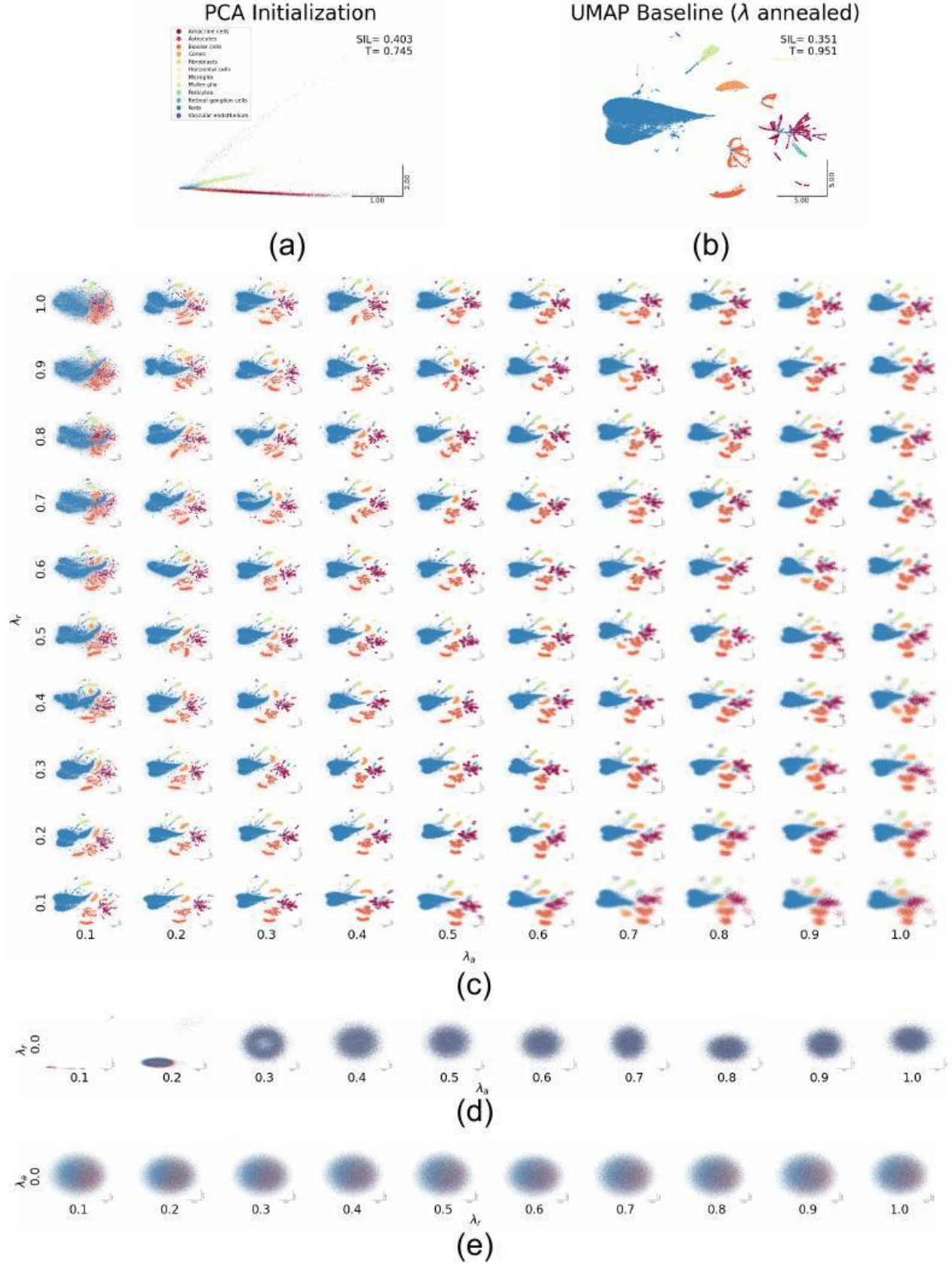


Figure 29: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

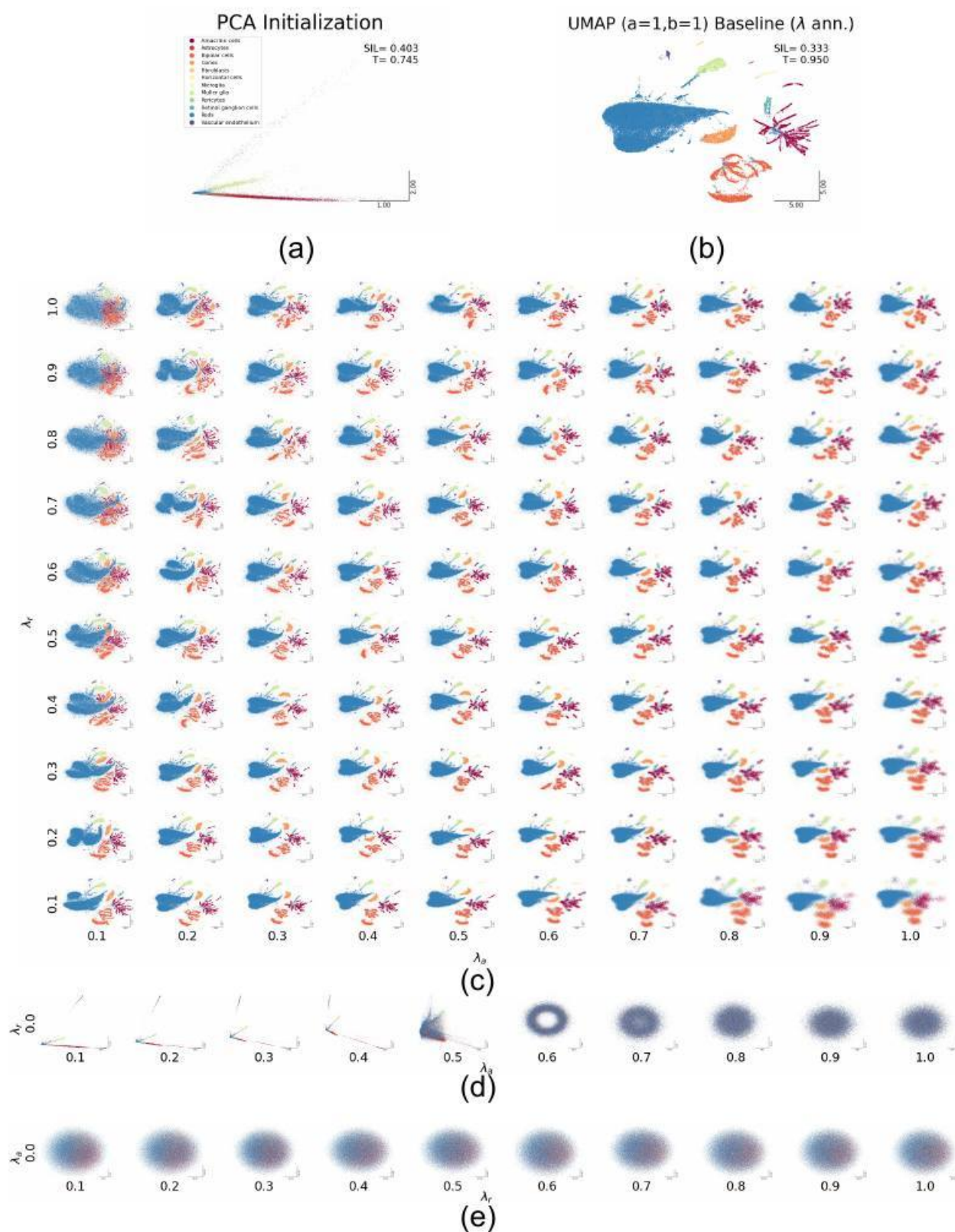


Figure 30: Varying λ_a and λ_r for the single-cell transcriptomes dataset using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), repulsive force alone cannot produce any clusters.

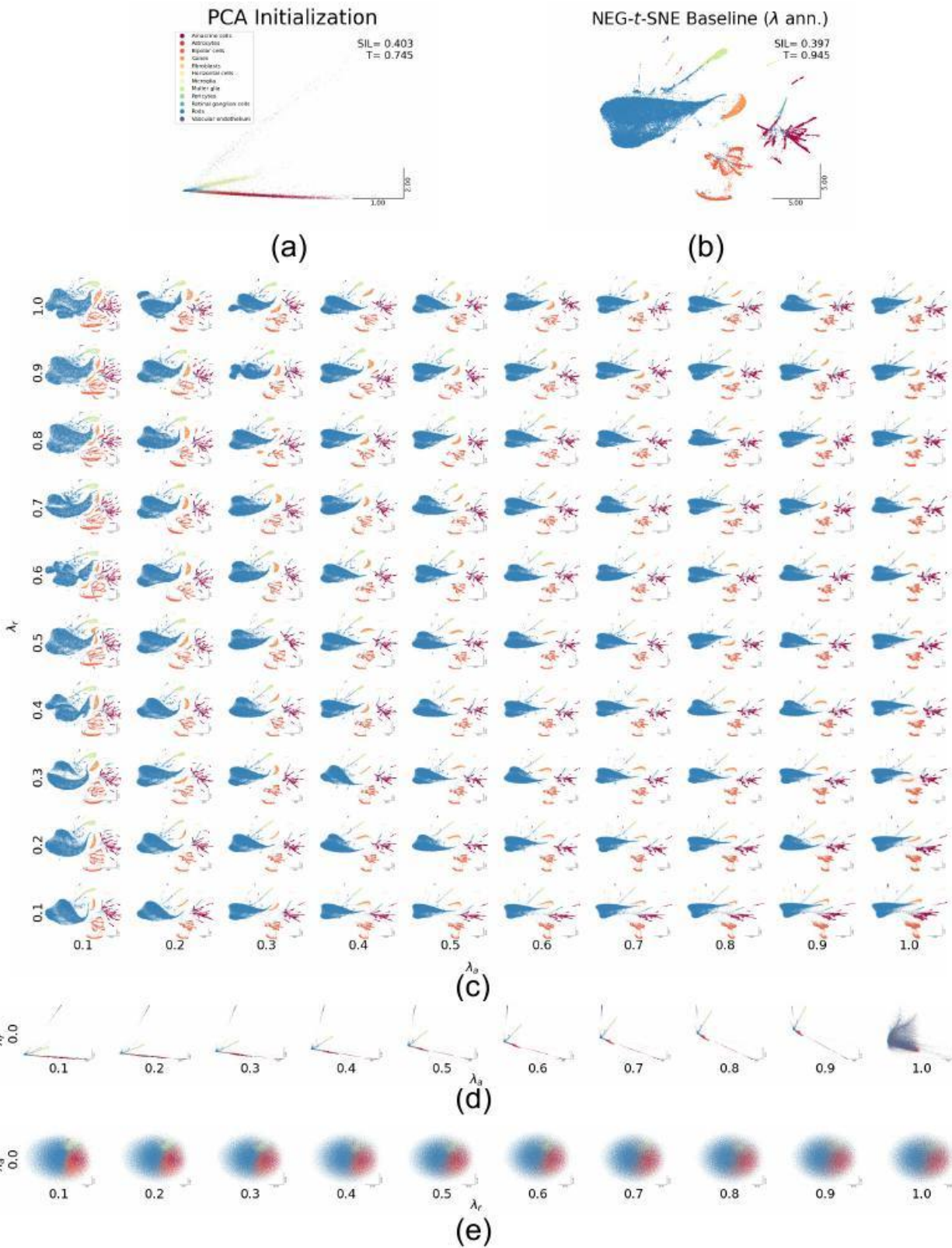


Figure 31: Varying λ_a and λ_r for the single-cell transcriptomes dataset using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no distinct clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

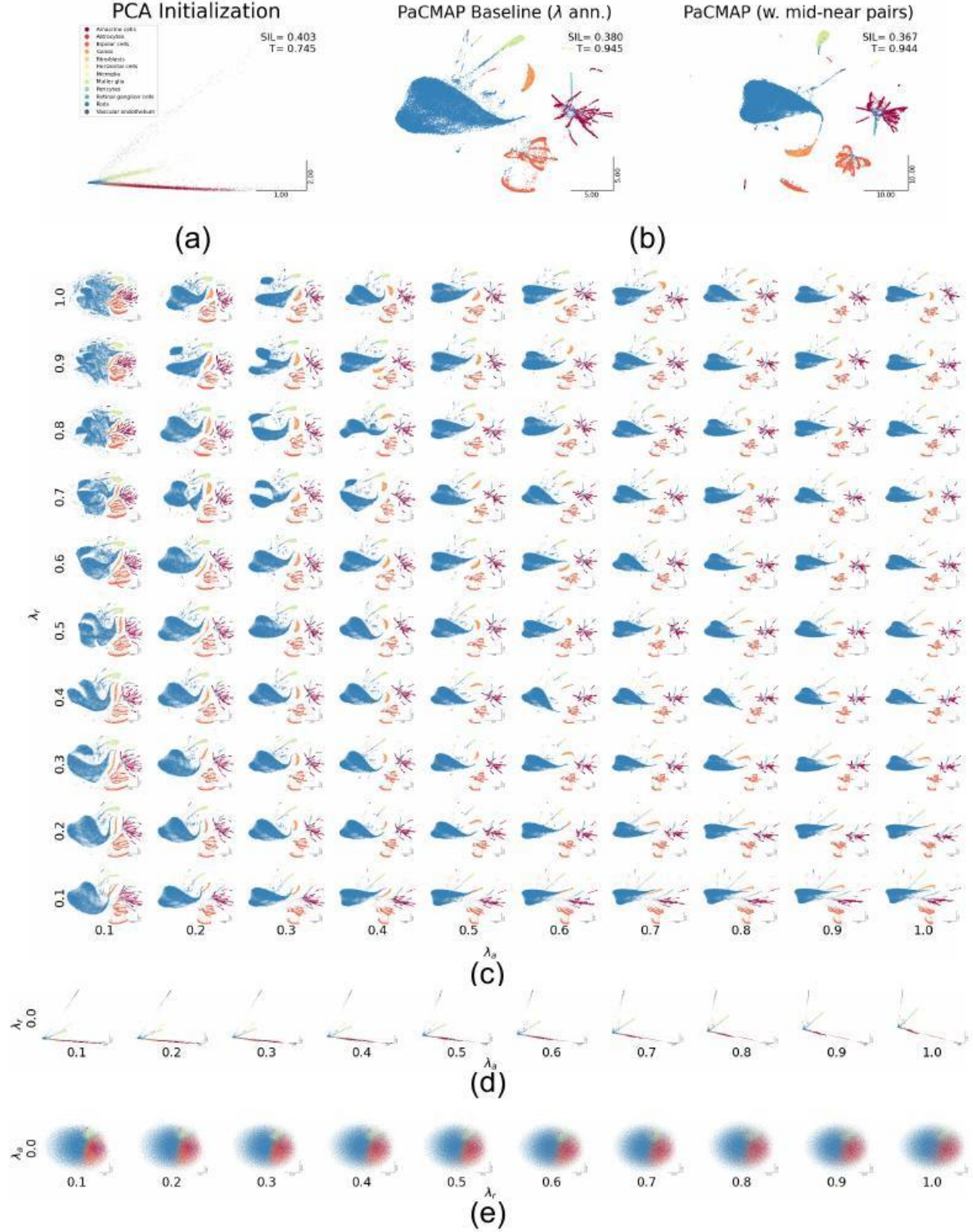


Figure 32: Varying λ_a and λ_r for the single-cell transcriptomes dataset using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) Embeddings as λ_a and λ_r vary (no annealing). (d) With $\lambda_r = 0$ (no repulsion), attraction alone forms no cluster. (e) With $\lambda_a = 0$ (no attraction), repulsion alone forms no cluster—together underscoring the attraction-repulsion division of labor of Contribution 4.

J Mix and Match Approach

A direct corollary of Contribution 1—that attraction and repulsion shapes are independent objects—is that shapes from different loss functions can be freely combined. Figure 33 shows several such combinations. To make the comparison fair, we restricted every attraction shape to $[-1, 0]$ (so Proposition 4.1 holds and learning-rate sensitivity is reduced) and used finite repulsion shapes (to curb large repulsive forces at small distances), holding the learning rate constant ($\lambda = 1$) and running the same UMAP optimization throughout so that any differences come from the shapes alone. The results are informative: UMAP’s attraction shape ($f_a = 0.5f_a^U$) gives better clusters (higher silhouette score), whereas PaCMAP’s gives better local structure (higher trustworthiness)—the latter because PaCMAP’s attraction sits within $[-0.5, 0]$ and so never flips pairs. Confining UMAP’s and NEG-*t*-SNE’s attraction to $[-0.5, 0]$ (via $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$) trades clustering performance for improved local structure (Fig. 34).

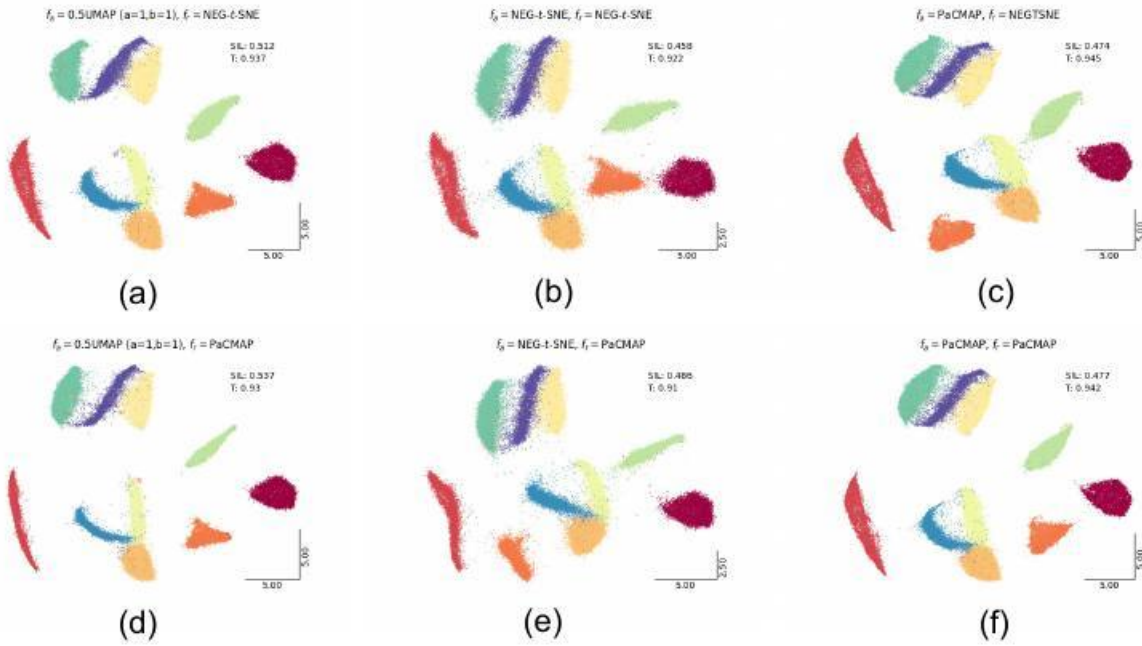


Figure 33: Mixing and matching attraction and repulsion shapes on MNIST at constant $\lambda = 1$ —a corollary of Contribution 1. Attraction shapes obey Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE’s repulsion. Bottom row: the same attraction shapes paired with PaCMAP’s repulsion.

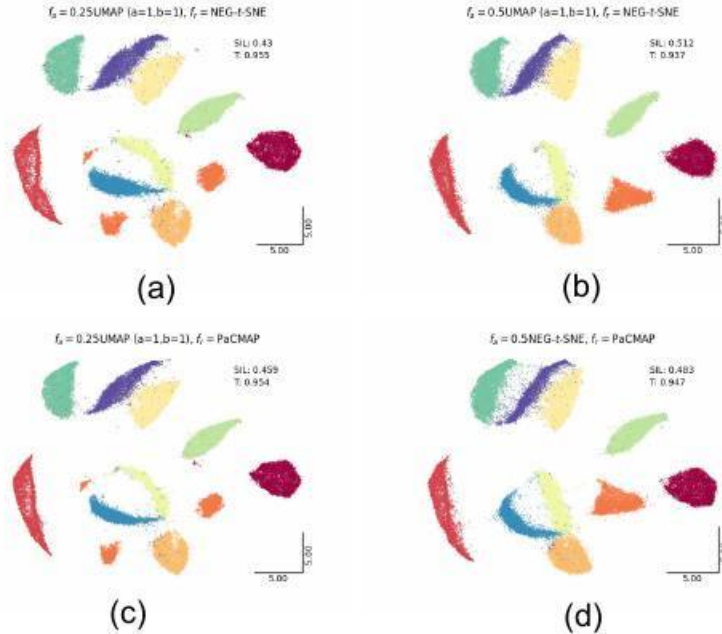


Figure 34: Mixing and matching shapes on MNIST at constant $\lambda = 1$, with attraction confined to $[-0.5, 0]$ to prevent flips during the attractive update. Top row: UMAP and NEG- t -SNE attraction shapes with NEG- t -SNE’s repulsion. Bottom row: the same attraction shapes with PaCMAP’s repulsion.

K Implementation Details

To keep the analysis controlled, we implemented our own UMAP so that the shapes could be swapped in isolation. We used numba (Lam et al. 2015) to build an exact (not approximate) nearest-neighbor graph with $k = 15$ and scikit-learn’s (Pedregosa et al. 2011) PCA for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9, and 10 and—by substituting the alternative methods’ attraction and repulsion shapes—the embeddings in Figs. 18, 34 (unless stated otherwise). Trustworthiness and silhouette scores use the corresponding scikit-learn functions.

The mappings in Figs. 2, 3, 9, 10, and 15 are rotated to a reference embedding (panel (a) of each figure). We did this by Procrustes alignment: normalizing each embedding to zero mean and unit norm, then extracting rotation and scaling with SciPy’s (Virtanen et al. 2020) `orthogonal_procrustes`.

To compare with Neg- t -SNE in Fig. 11, we used the original implementation of the contrastive embedding framework for both the UMAP and the Neg- t -SNE algorithms; the implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22(b) (right), 27(b) (right), and 32(b) (right) were obtained using the official PaCMAP package; the implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.